

筑波大講義

統計神経力学

ーランダム結合回路網の巨視的力学

甘利俊一

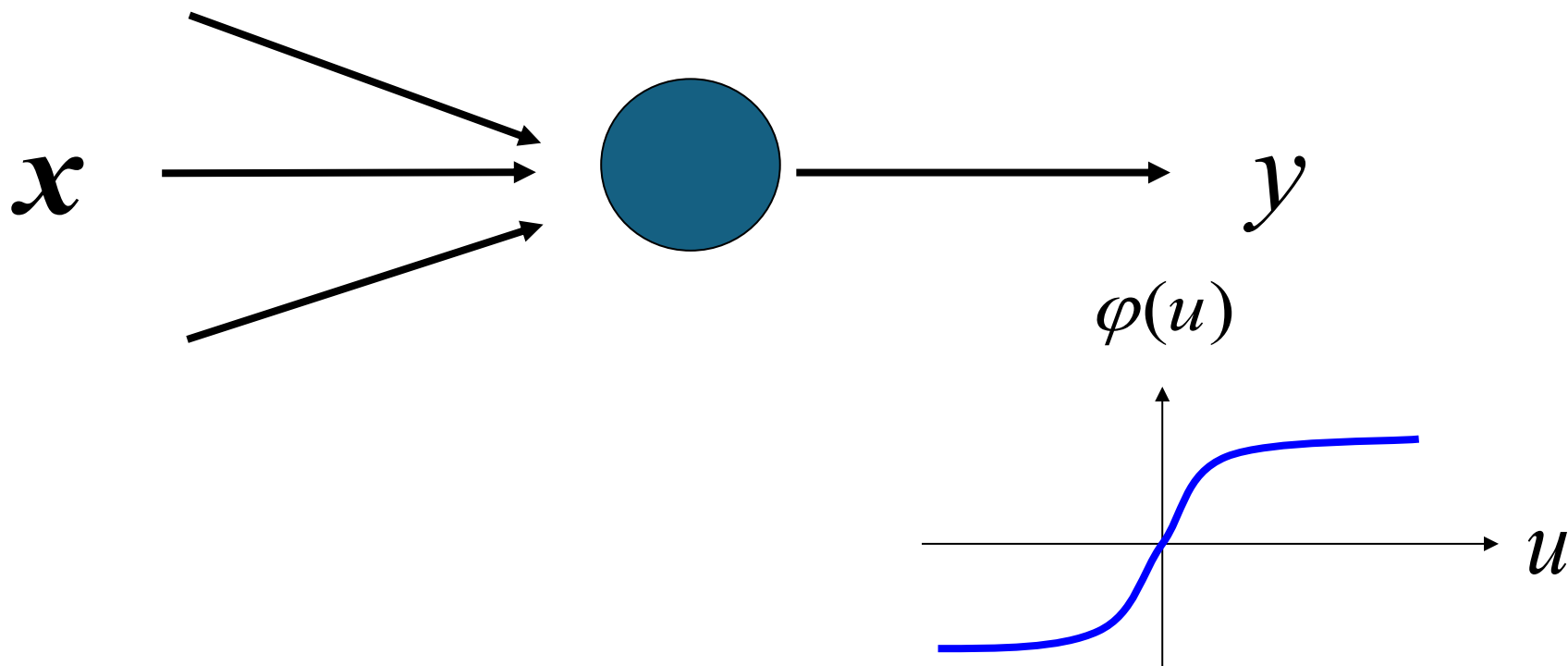
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数理ニューロン

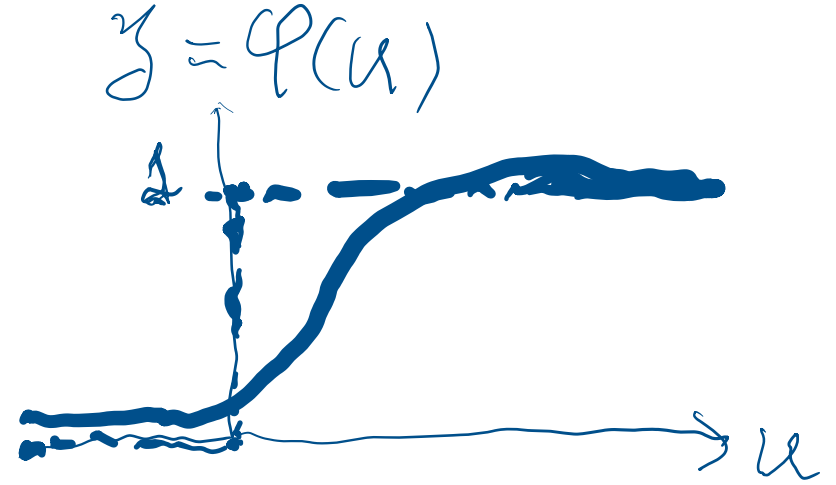
$$y = \varphi\left(\sum w_i x_i - h\right) = \varphi(\boldsymbol{w} \cdot \boldsymbol{x})$$



神経細胞の数理モデル：ランダム結合

$$y = \varphi(w \cdot x - b)$$

$$u = w \cdot x = \sum_{i=1}^n w_i x_i - b$$



w_i, b : Random; 独立

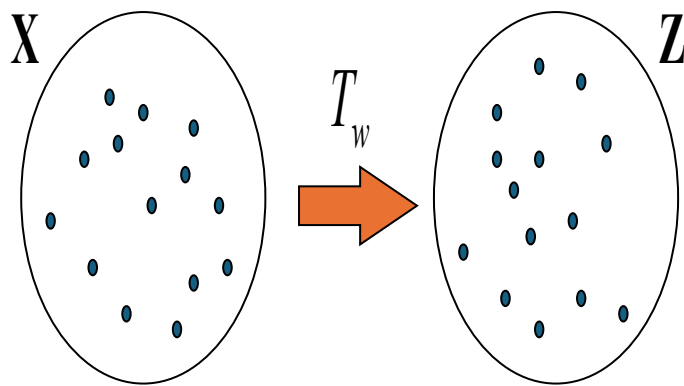
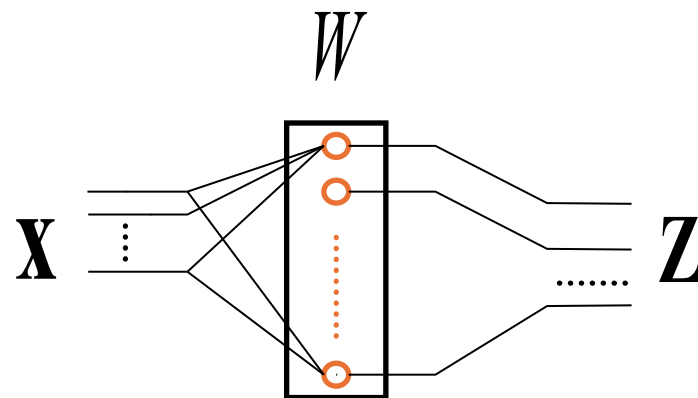
$$V[u] = \sigma^2 + \sigma_b^2 \sim O(1)$$

$$N(0, \frac{\sigma^2}{n})$$

$$b \sim N(0, \sigma_b^2)$$

層状の神経回路

$$\mathbf{z} = T_w \mathbf{x}$$

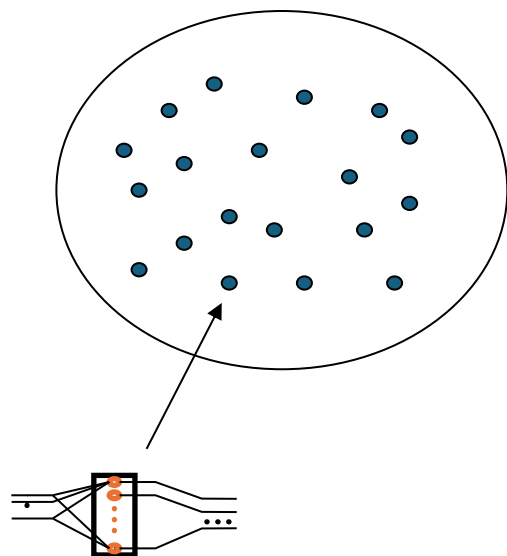
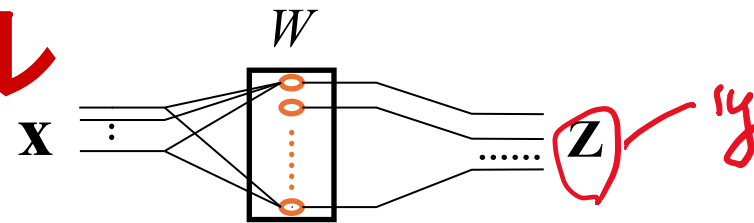


$$z_i = \varphi \left(\sum_j w_{ij} x_j - b_i \right)$$

$$\mathbf{z} = \varphi (W \mathbf{x} - \mathbf{b})$$

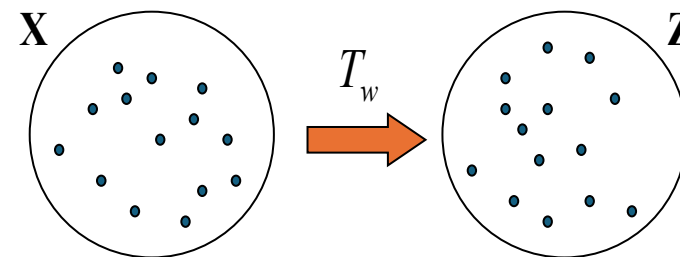
層状神経回路網のアンサンブル

$$\Omega = \{w\}, \quad p(w)$$



microdynamics

$$\underline{\mathbf{z}} = \underline{T_w \mathbf{x}}$$



macrostate

$$X = X(\mathbf{x}); \quad Z = Z(\mathbf{z})$$

$$X = \frac{1}{n} \sum x_i, \quad Z = \frac{1}{n} \sum z_i$$

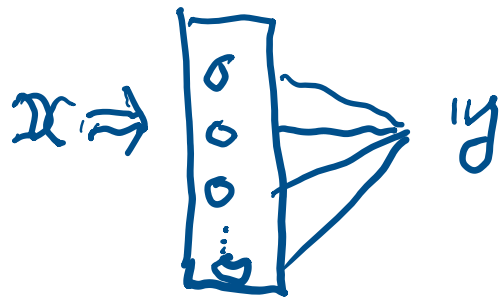
$$Z = Z(\mathbf{z}) = Z(T_w \mathbf{x})$$

$$Z = F(X(\mathbf{x}))$$

$$\underline{Z = F(X)} : \text{Macroscopic law}$$

: ほとんどの状態の回路から

1層の回路網と巨視的量



$$y_i = \varphi(w_i \cdot x - b_i), \quad y = \varphi(Wx - b)$$

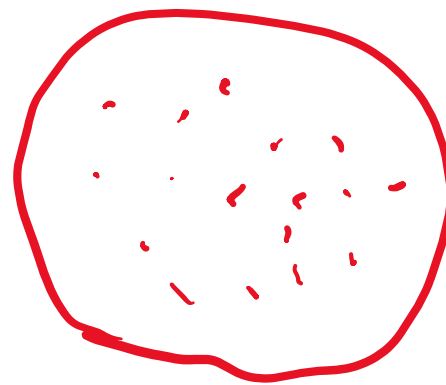
$$\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{pmatrix}$$

$$A_x = \frac{1}{n} \sum x_i^2, \quad A_y = \frac{1}{m} \sum y_i^2$$

活動度

$$A_y = \hat{F}(A_x)$$

巨視法則



$$n, m \rightarrow \infty$$

$$p \rightarrow 1$$

活動度の巨視法則の導出

$$A_Y = E[\varphi(u_i)^2], \quad u_i = \sum_j w_{ij} x_j - b_i$$

$$p(u_i) \sim N(\tilde{b}, \sigma^2 A_x + \sigma_h^2)$$

$$A_Y = \int \varphi(u_i)^2 p(u_i) du_i$$

$$\approx F(A_x)$$



新しい巨視量：信号変換の距離法則

$$D_x = \frac{1}{n} \sum |x_i - x'_i|^2$$

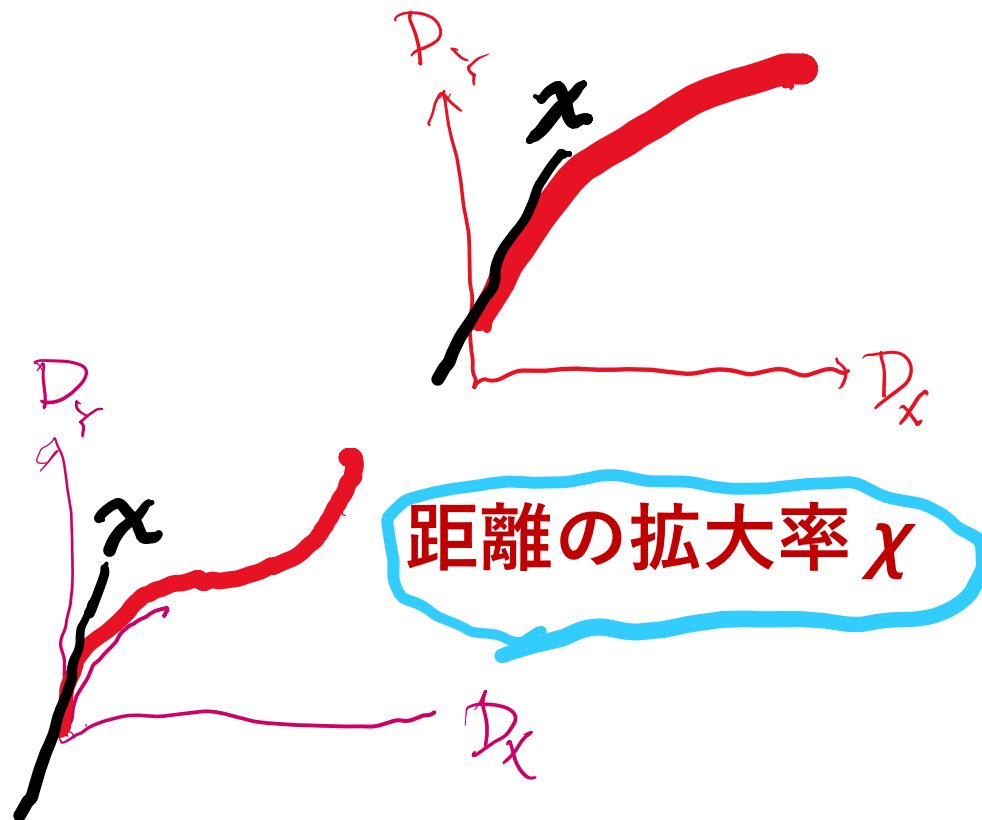
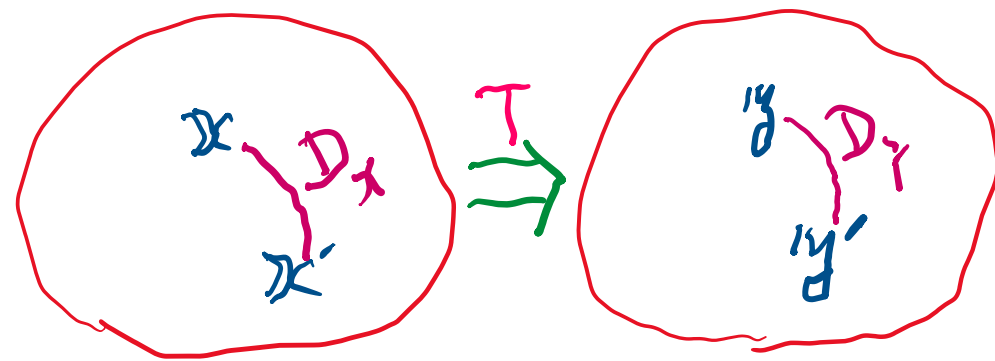
$$D_y = \frac{1}{m} \sum |y_i - y'_i|^2$$

$$D_y = \hat{\Phi}_D(D_x, A_x)$$

2値ニューロン 1, -1

$$D_y = \frac{2}{\pi} \sin^{-1} \sqrt{\frac{\sigma_w^2}{\sigma_w^2 + \sigma_h^2}} D_x$$

演習問題



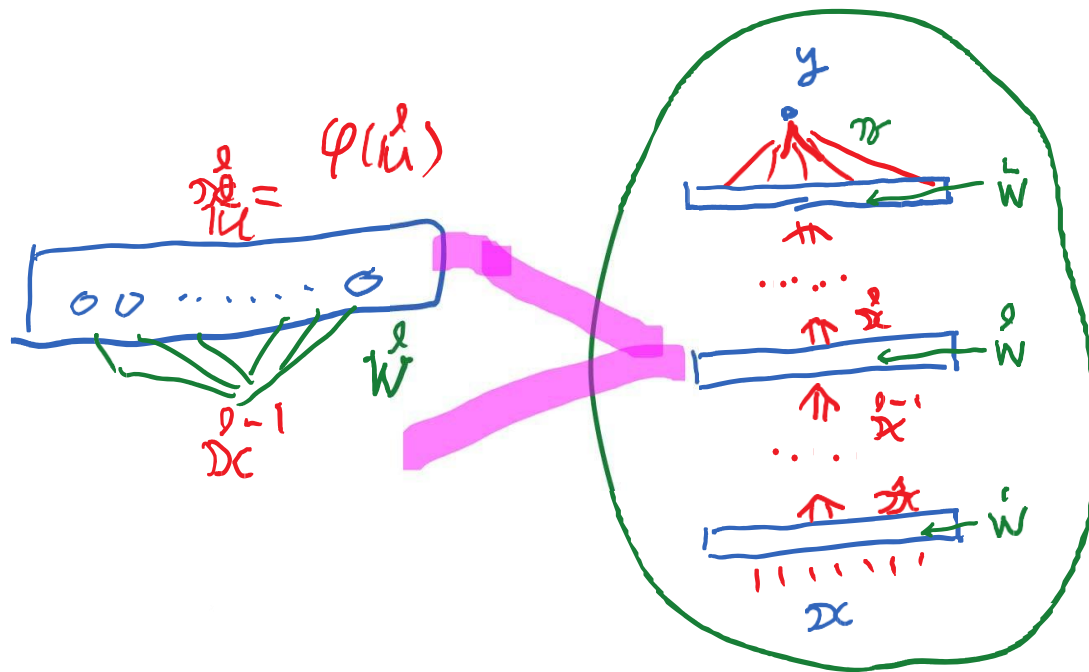
深層回路網

$$y = \sum_i^L w_i^L x_i$$

$$x_i = \varphi(u_i) = \varphi(\mathbf{u})$$

$$u_i = \sum_j^{l-1} w_{ij}^l x_j - h_i$$

$$\theta = (w_i^L, w_{ij}^{L-1}, \dots, w_{ij}^1, h)$$



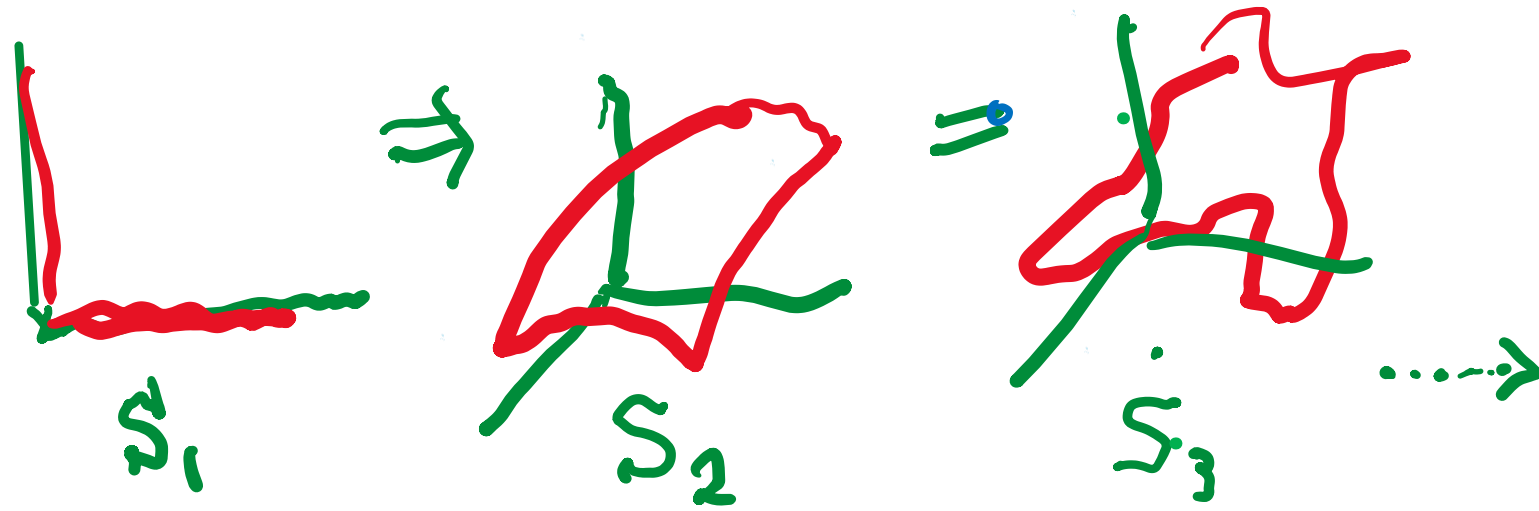
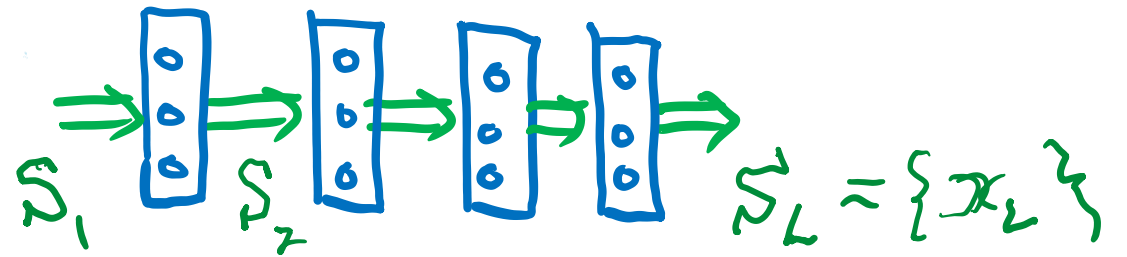
$$y = f(x, \theta)$$

巨視的ダイナミクス

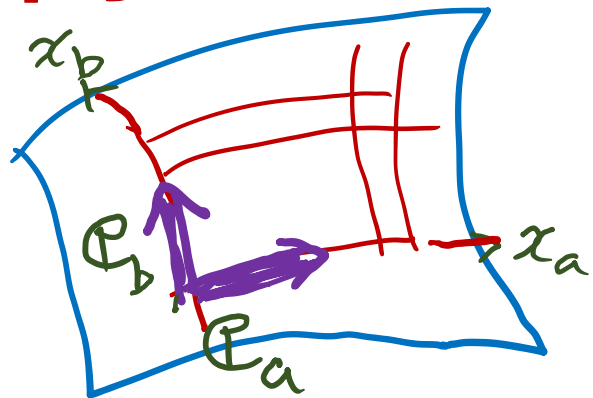
——活動度、計量、曲率 ... 距離

$$A_{t+r} = \tilde{\Phi}_A(A_t)$$

$$D_{t+r} = \tilde{\Phi}_D(D_t)$$



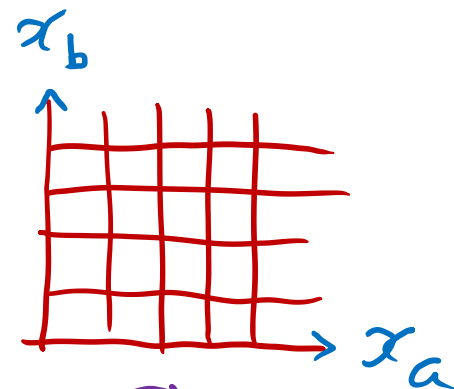
接空間



$$g_{ab} = \frac{1}{n_l} \mathbf{e}_a \cdot \mathbf{e}_b$$

計量：長と角度

$$T \mathbf{e}_a' = B \mathbf{e}_a$$



$T \ll S_0$

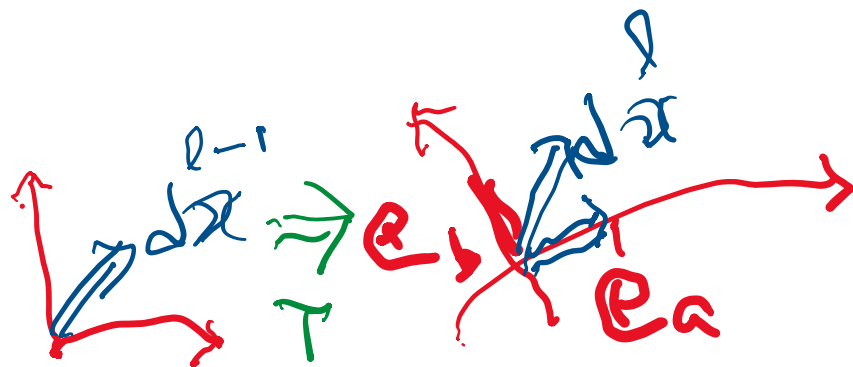


計量 metric

$$dx = \sum dx^a e_a$$

$$dx^0 = \frac{\partial x^0}{\partial x^{l-1}} dx^{l-1} = \underline{\dot{X}} dx^{l-1} ; \quad \dot{X} = \varphi'(W \dot{x}^{l-1})$$

$$\dot{X} = D^l W^l ; \quad D = \text{diag}(\varphi'(u))$$



計量の変換

$$dS^2 = \left| \sum^p X d\tilde{x}^{p-1} \right|^2 = d\tilde{x}^{p-1} \left(\sum^p X^T X \right) d\tilde{x}^{p-1}$$

$$\sum^p X = \frac{\partial \sum^p x}{\partial \tilde{x}^{p-1}}$$

$$X^T X = E \left[\left\{ \varphi'(u_i) \right\}^2 w_{ij} w_{ik} \right]$$

$$\underbrace{dS^2 = \sum^p X_1 d\tilde{x}^{p-1}}_{\substack{\sum^p \\ X_1}} \quad E[w_{ij} w_{ik}] = \delta_{ik} \frac{\sigma^2}{p}$$

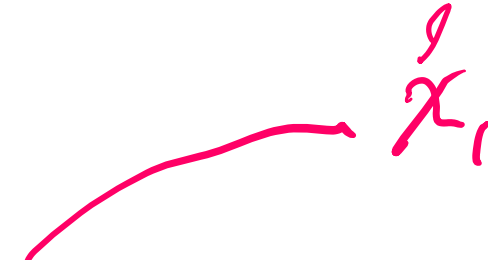
$$dS^2 = \int \tilde{g}_{ab} d\tilde{x}^a d\tilde{x}^b = d\tilde{x}^a \tilde{G}_a d\tilde{x}^a$$

Metric

Law of large numbers

$$g_{ab}^l = \left\langle \mathbf{e}_a^l, \mathbf{e}_b^l \right\rangle = BB^l g_{ab}^{l-1}$$

$$ds^2 = \sum_{a,b} g_{ab}^l dx_a^l dx_b^l$$

$$BB^l = \sum_{i_l} w_{i_{l-1}}^{i_l} w_{i'_{l-1}}^{i_l} \varphi'(u_{i_l})^2 \approx \sigma_l^2 E[\varphi'^2] \delta_{i_{l-1} i'_{l-1}}$$


$$\chi_1^l = \sigma_l^2 E \left[\varphi' \left(\tilde{u}_{i_l}^l \right)^2 \right]$$

$$\tilde{g}_{ij}(x) = \chi_1(A) g_{ij}(x)$$

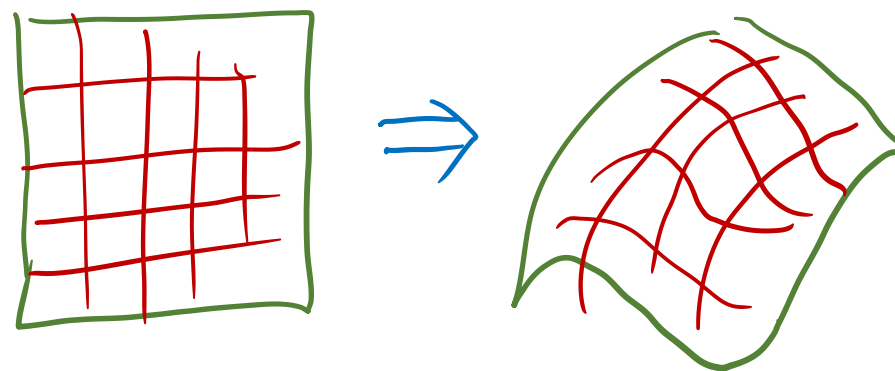
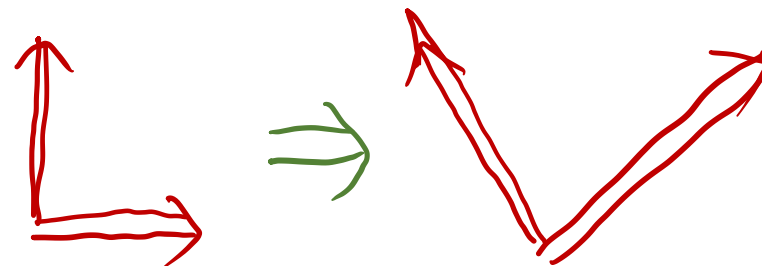
conformal transformation!

$$\bar{\chi}_1 = \bar{\chi}_1(\bar{A}) \delta_{ij}$$

$$\Rightarrow g_{ij} = \bar{\chi}_1^l \delta_{ij}$$



回転と拡大

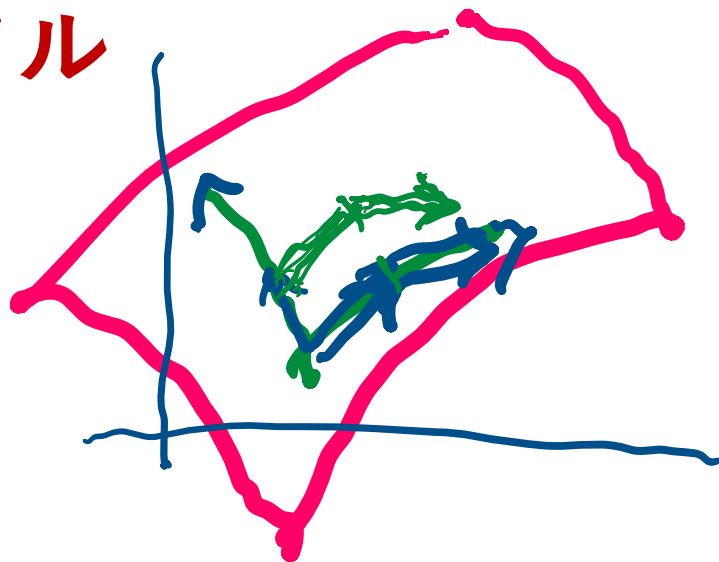


曲率の力学

Euler-Schouten 曲率テンソル (挿入曲率)

$$\rho_{ab}^0$$

γ : 曲率

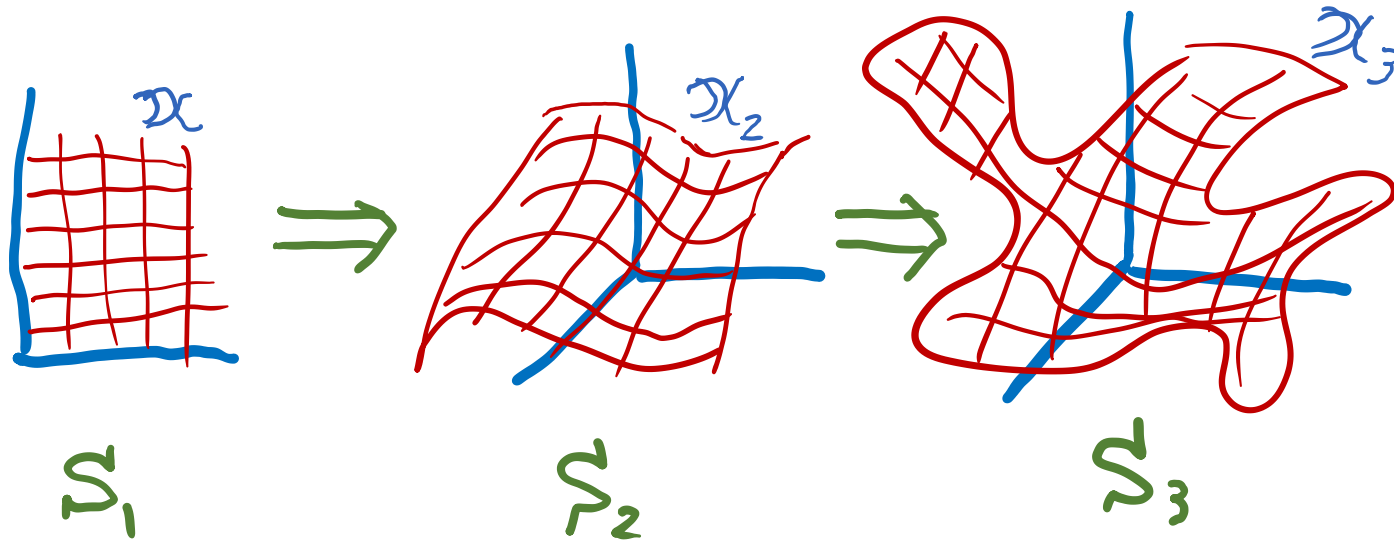


$$\gamma^2 = \frac{1}{\chi_1^{l-1}} \gamma^{l-1} + \frac{3}{p_0 \chi_1^{l-1}} \chi_2^{l-1}$$

$\sigma^2 E[\{\varphi''(\alpha)\}^2]$

$$\gamma^2 = \frac{3\chi_2}{p \pi \chi_1^k (\pi \chi_1^k - 1)}$$

計量と曲率



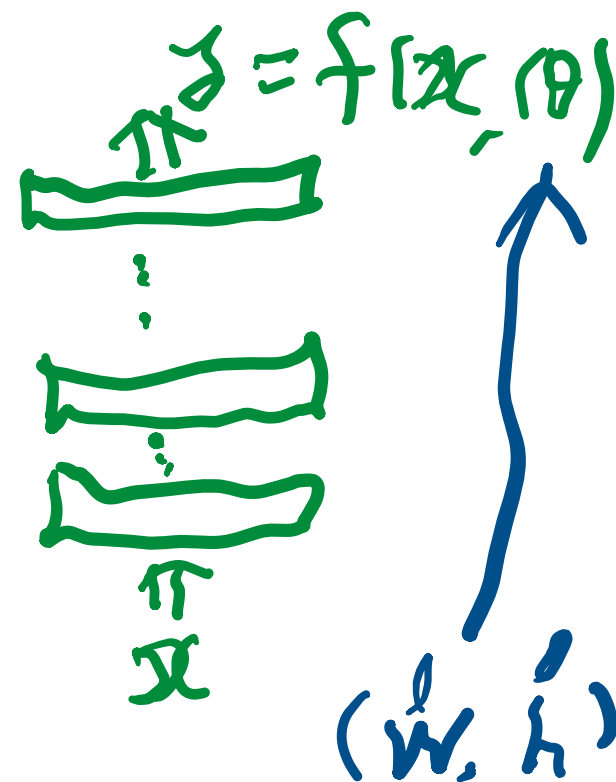
$$\tilde{x} = \phi(Wx)$$

$$ds^2 = \sum^l g_{ij} dx^i dx^j = \frac{1}{n_l} d\mathbf{x} \cdot d\mathbf{x}$$

神経回路網の普遍近似定理 と曲率

神経回路網の普遍近似定理

いかなる関数 $f(x)$ も $f(x, \theta)$ で十分に
近似できる : $p \rightarrow \infty$

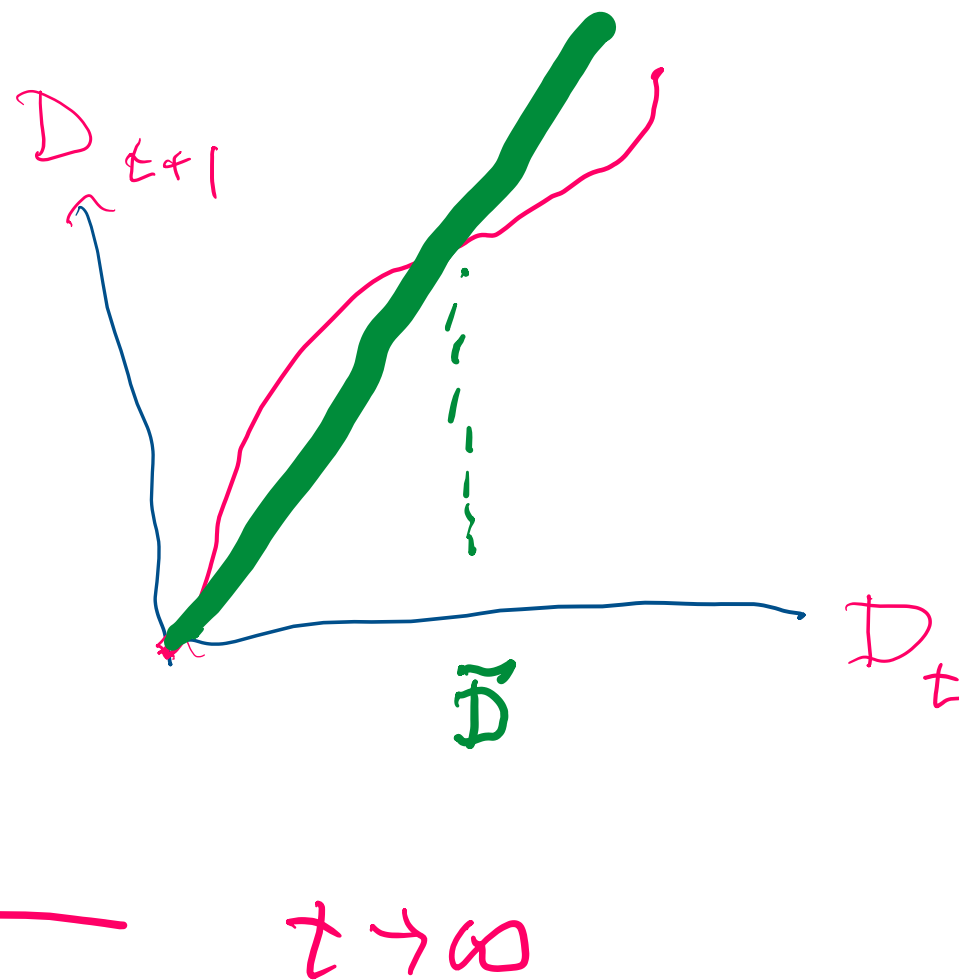


距離法則の巨視的力学

$$D_{t+1} \approx \Phi_D(D_t)$$

$$x_i = \pm 1.$$

$$D_{t+1} = \frac{2}{\pi} \sin^{-1} \sqrt{c D_t}$$



巨視的法則は成立するのか？

$$\mathcal{D}_{t+1} = \bar{\Phi}_{\mathcal{D}}(\mathcal{D}_t)$$

$$x_i^t = \text{sgn}\left(\sum w_{ij}x_j^{t-1} - h_i\right) \quad x_i = \pm 1$$

$t \rightarrow \infty$

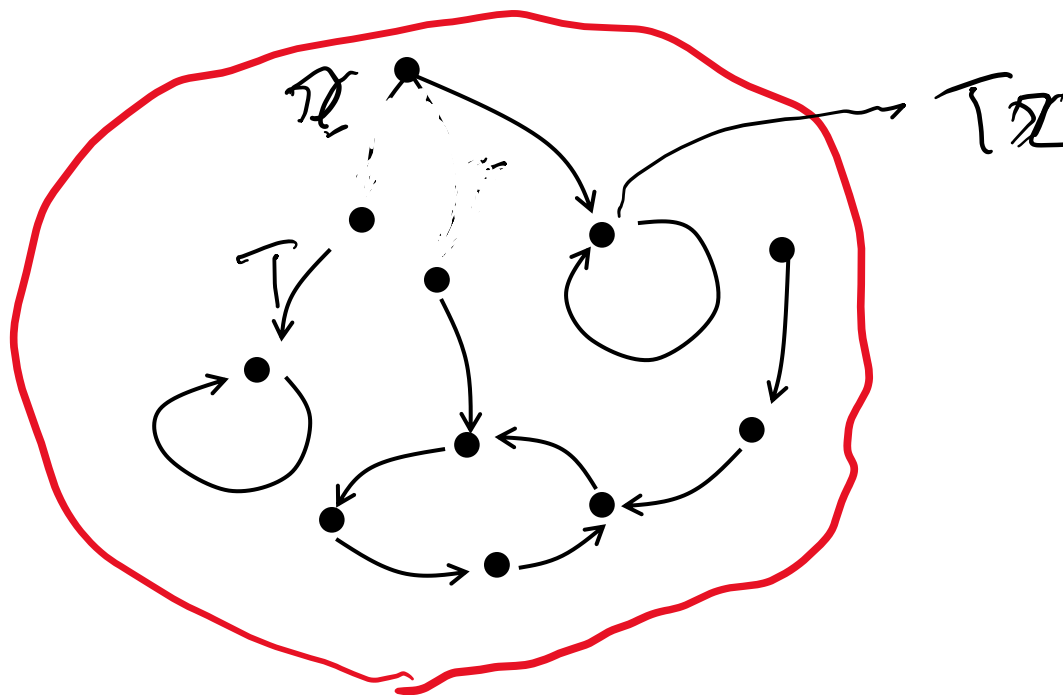
$$\mathbf{x}^t = \text{sgn}(W\mathbf{x}^{t-1} - \mathbf{h}) = T_{\omega}\mathbf{x}^{t-1}$$

$$X' = F(X) = F(X(T_{\omega}(\mathbf{x})))$$

$$X_{t+1} = F(X_t)? \quad F(X(T_{\omega}T_{\omega}(\mathbf{x})))$$

$$\omega = (W, \mathbf{h})$$

状態遷移図



距離の神経力学

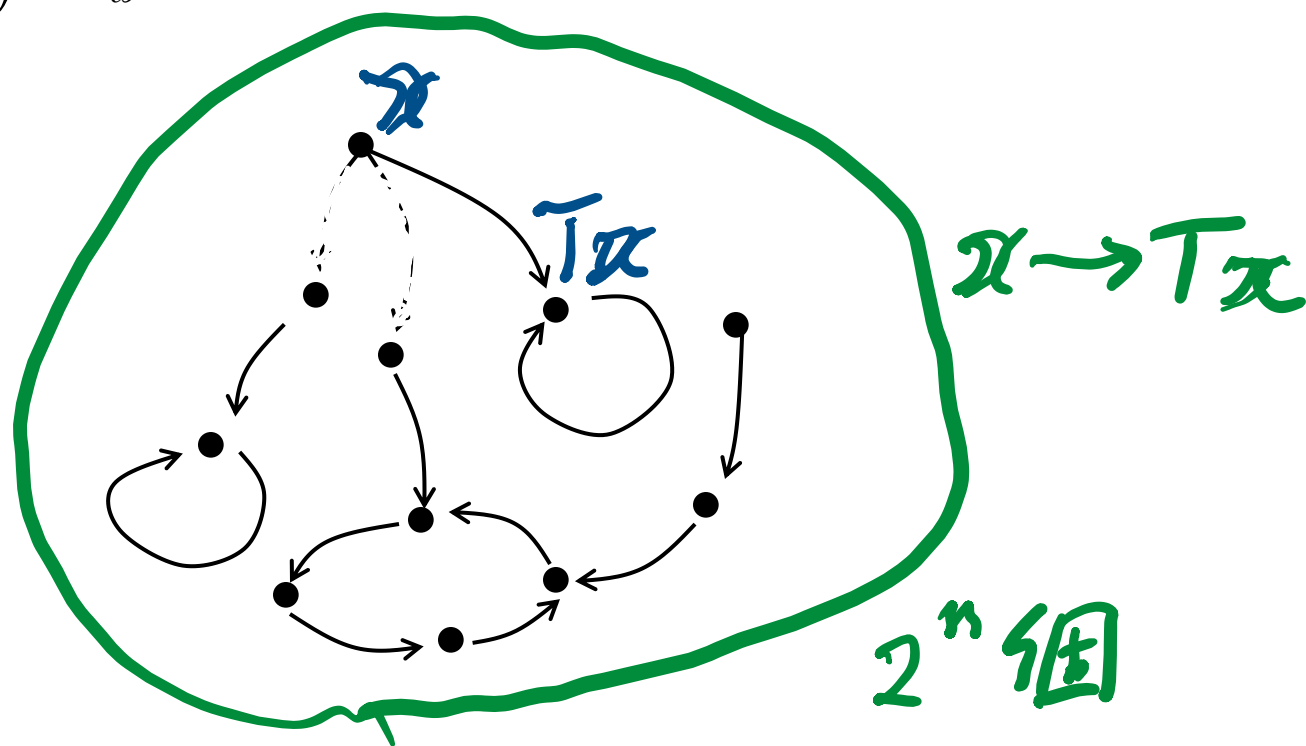
$$D_{t+1} = \bar{\Phi}(D_t)$$

$$x_i^t = \text{sgn}\left(\sum w_{ij} x_j^{t-1} - h_i\right) \quad x_i = \pm 1$$

$$\mathbf{x}^t = \text{sgn}\left(W\mathbf{x}^{t-1} - \mathbf{h}\right) = T_{\omega}\mathbf{x}^{t-1}$$

$$\omega = (W, \mathbf{h})$$

状態遷移図



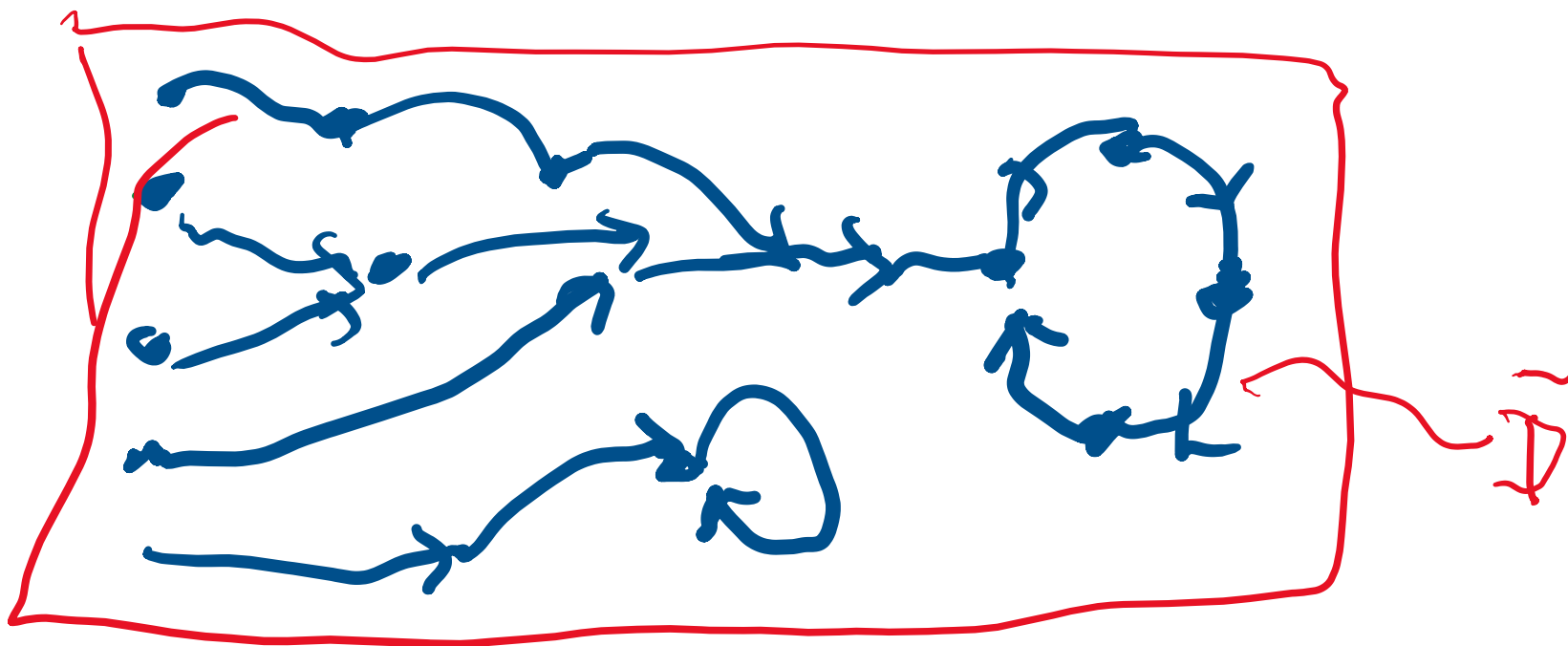
離散ニューロン回路とアトラクター

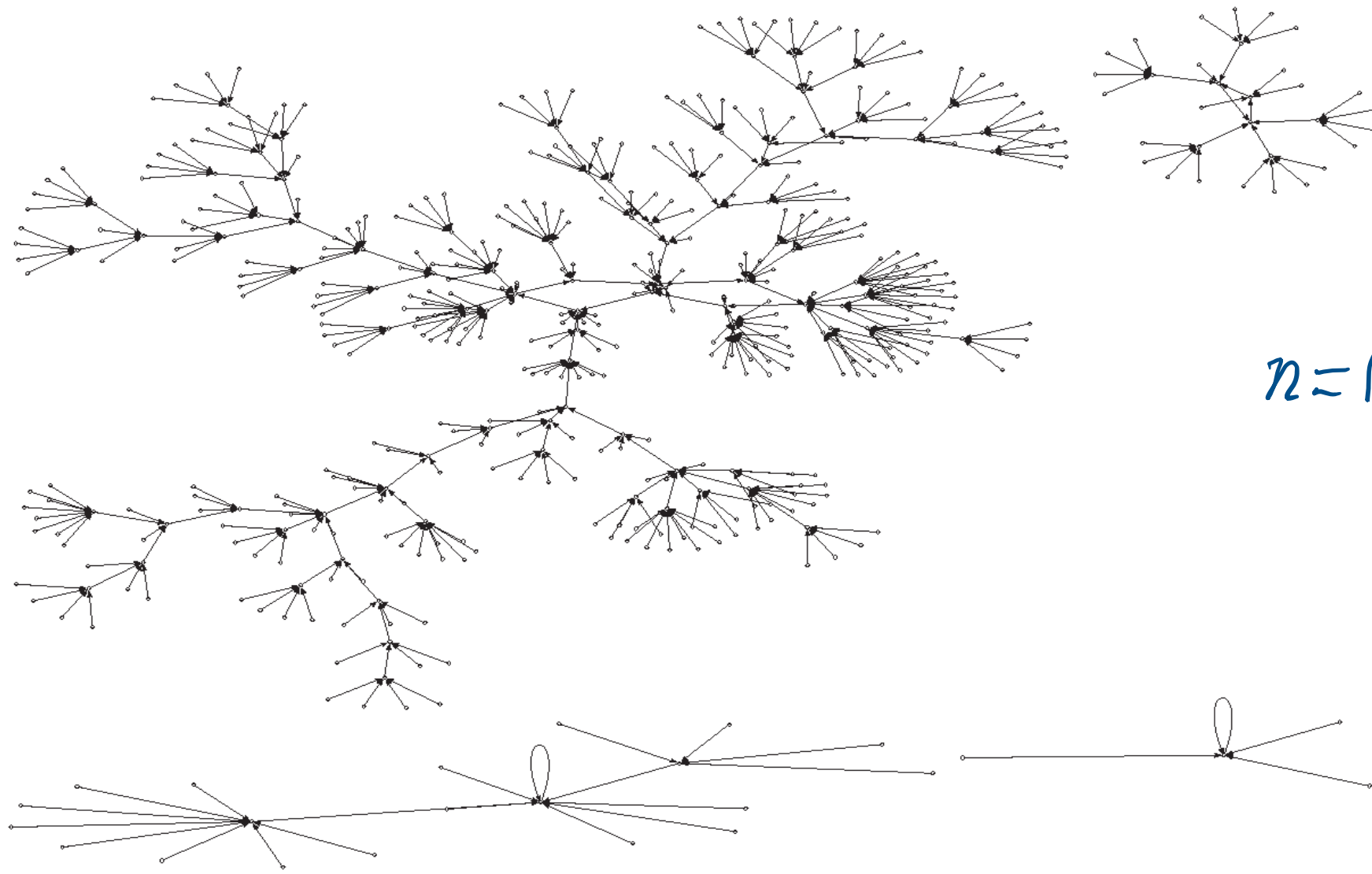
破綻

状態遷移図

$$D_{t+1} = \frac{2}{\pi} \sin^{-1} \sqrt{D_t}$$

アトラクター数
リミットサイクルの周期



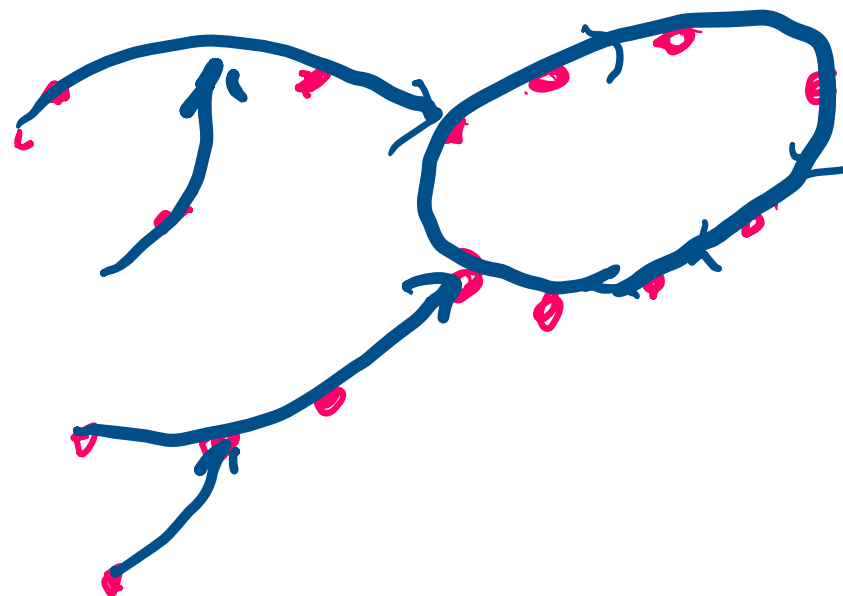


$$n=13$$

アトラクター リミットサイクル 周期

アトラクターに落ちるまでの長さ

ランダムグラフとの対比



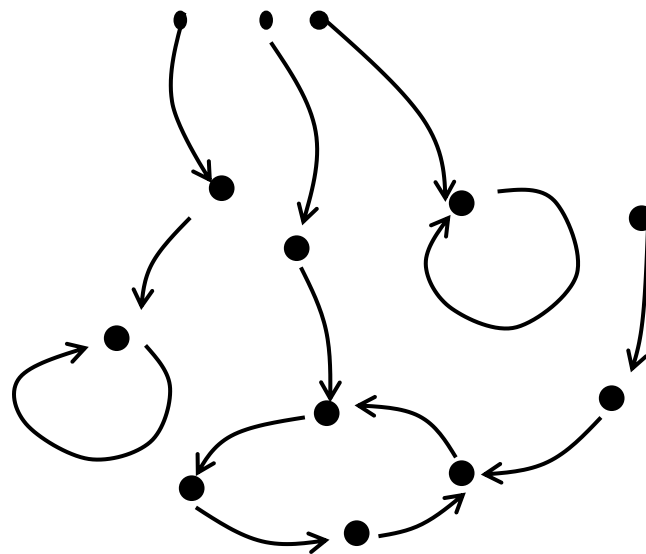
状態遷移図

$$p_k = \text{Prob} \left\{ |T_{\omega}^{-1} \mathbf{x}| = k \right\}$$

$$\sum p_k = 1, \quad \sum k p_k = 1,$$

$$c = \sum k^2 p_k - 1 : \text{variance}$$

$$\text{neural} : \quad c_n \rightarrow \infty \quad ; \quad \text{random graph} : c \rightarrow 1$$



先祖の数の独占： 消失する状態数大、過渡状態短

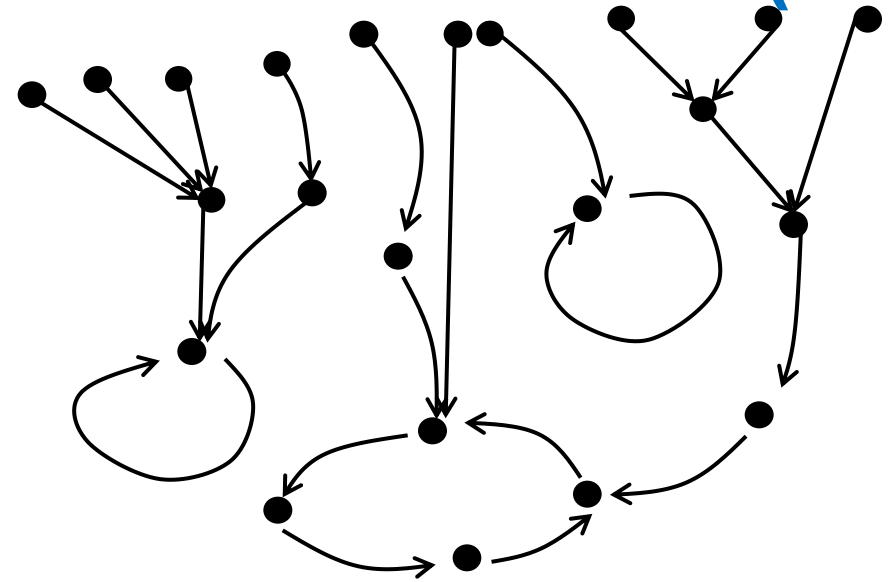
収束の速さ Concentration of wealth (monopoly)

$$\mathbf{s}_0 = (s_{0\alpha}) = (1, 1, 1, \dots, 1)$$

$$T_\omega \mathbf{x}_\beta = \mathbf{x}_\alpha : \mathbf{x}_\beta \rightarrow \mathbf{x}_\alpha : T_{\alpha\beta}$$

$$\mathbf{s}_1 = T_\omega \mathbf{s}_0; \quad \mathbf{s}_t = T_\omega^t \mathbf{s}_0$$

$$\bar{S}_1 = E[s_{1\alpha} \mid \mathbf{x}_\beta \rightarrow \mathbf{x}_\alpha] : \quad \bar{S}_t = E[s_{1\alpha} \mid \mathbf{x}_\beta \rightarrow \mathbf{x}_{\beta'} \dots \rightarrow \mathbf{x}_\alpha]$$



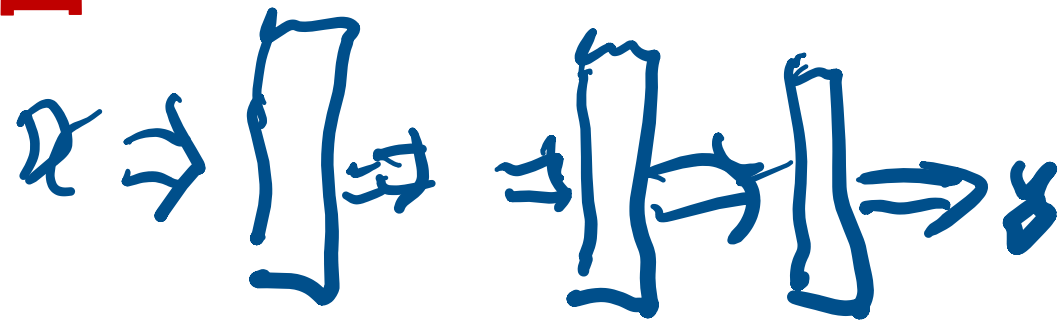
**Expected number of in-coming branches
when $\mathbf{x}$ is not in the garden of Eden**

弱法則 OK; 強法則は成立しない

距離の統計神経力学

弱法則 OK;
強法則は成立しない

深層神経回路網の学習



$$y = f(x, \theta) = \varphi(\tilde{w}^L \varphi(\tilde{w}^{L-1} \varphi(\dots)))$$

訓練データ

$$D = \{(x_1, y_1), \dots, (x_n, y_n)\}$$

$$y_t \approx f(x_t, \theta_t) + \varepsilon_t \sim N(0, 1)$$

正解

$$e_t = \frac{1}{2} (y_t^* - y_t)^2 \approx \frac{1}{2} \{ f(x_t, \theta^*) - f(x_t, \theta_t) \}^2$$

パラメータ空間

確率分布のなす空間

$$y = \varphi(Wx\varphi(Wx\dots\dots)) + \varepsilon; \quad \varepsilon \sim N(0,1)$$

$$p(y, x : \theta) = c \exp \left\{ -\frac{1}{2} (y - f(x; \theta))^2 \right\} q(x)$$

$$G = E_x [\nabla_W \log p(y, x : \theta) \nabla_\theta \log p(y, x : W)]$$

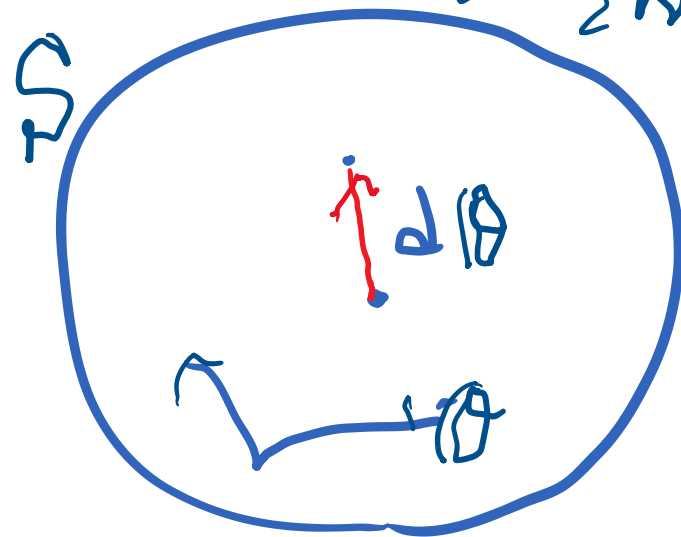
$$ds^2 = d\theta^T G d\theta$$

Fisher information

error

$$\mathcal{S} = \{\theta\} \sim \{\vec{w} \dots \vec{w}, h \dots\}$$

$$\theta \sim \{W\}$$



Riemannian

確率の勾配降下学習法 — Amari, 1967)

$$l_1 = \frac{1}{2} e_t^2$$

$$L(\theta) \approx E[l(x, y, \theta)]$$

$$\Delta \theta_t \approx -\eta_t \frac{\partial l(x_t, y_t, \theta_t)}{\partial \theta}$$

$$l = \frac{1}{2} e^2$$

$$e = y - f(x, \theta)$$

$$= -\eta_t e_t \frac{\partial f(x_t, \theta_t)}{\partial \theta}$$



誤差逆伝搬法

$$\theta = \{\tilde{w}, \tilde{w}', \dots, \tilde{w}^L\}$$

$$\Delta \tilde{w} = -\eta \frac{\partial \mathcal{F}}{\partial \tilde{w}}$$

$$\lambda_{l+1}^i = \frac{\partial x_{l+1}^i}{\partial x_l^i}, \quad \lambda_{l+1}^j = \frac{\partial u_{l+1}^j}{\partial u_l^j}$$

$$\approx \begin{matrix} \lambda_{l+1}^i & \lambda_{l+1}^j \\ D & W \end{matrix}$$

$$\lambda_{l+1}^i = \frac{\partial x_{l+1}^i}{\partial u_l^j}$$

$$\begin{matrix} \lambda_{l+1}^i \\ W \end{matrix} D$$

$$x \Rightarrow [\] \Rightarrow [\] \dots [\] \rightarrow y$$

$$\mathcal{F}(\tilde{w} \varphi(\tilde{w}' \dots x))$$

$$W \tilde{x}^{-1} = \tilde{x}$$

$$D = \text{diag}[\varphi'(u^i)]$$

$$\frac{\partial f}{\partial \vec{w}} = \vec{U}^{L+1} \vec{U}^L \dots \vec{U}^0 \cdot \vec{x}^{0-1}$$

$$\vec{z} = \frac{\partial \phi}{\partial \vec{w}} = \underbrace{e \vec{U}^{L+1} \vec{U}^L \dots \vec{U}^0}_{\vec{e}} \cdot \vec{x}^{0-1}$$

$$\Delta \vec{w} = -\eta \vec{e} \vec{x}^{0-1} \quad ;$$

$$\underline{\vec{e}^{L+1} = e \vec{U}^L}$$

$$\hat{U} \hat{U}^\tau = \chi_s I ; \quad \chi \chi^\tau = \chi_i I$$

$$|e|^2 = \chi_i |e^{k+1}|^2 = \prod_k \chi_i e^2$$

$$\prod \chi_i \approx 1$$

カオスの淵

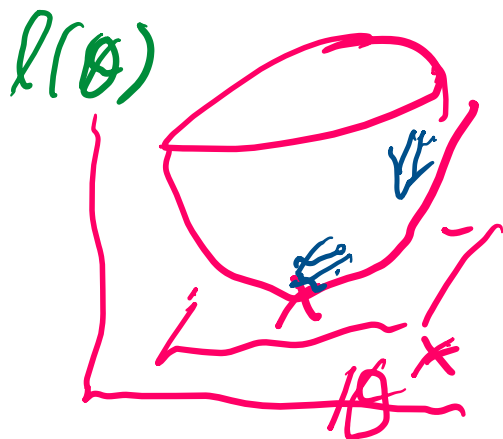
学習の速度と精度

θ^* : 最適点

$$\theta_{t+1} = \theta_t - \eta \frac{\partial l(x_t, y_t, \theta_t)}{\partial \theta}$$

$$\frac{\partial L(\theta^* + \Delta \theta)}{\partial \theta} = \frac{\partial L(\theta^*)}{\partial \theta} + \frac{\partial^2 L(\theta^*)}{\partial \theta \partial \theta} \Delta \theta$$

$\approx H$



θ^* の近傍での動作

ヘッシアンHの固有値分解

$$H = S \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_p \end{bmatrix} S^T$$

S : 直交行列

固有方向に分解

$$\Delta \theta_t^i = (1 - \eta \lambda_i) \Delta \theta_{t-1}^i + \varepsilon \quad i=1, \dots, p$$

$$-1 > 1 - \eta \lambda > 1$$

$$\eta < \frac{2}{\lambda_{\max}}$$

収束遅い!
これが大きいと

学習の加速 ニュートン法

$$\Delta \theta \approx -\eta H^{-1} \frac{\partial \ell}{\partial \theta}$$

$$H = F \text{ at } \theta^*$$

Fisher 情報行列

正定とは限らない

統計モデル：確率分布族の空間

$$\mathcal{S} = \{p(x, \theta)\}$$

$$y = \varphi(Wx\varphi(Wx\dots\dots)) + \varepsilon; \quad \varepsilon \sim N(0, 1)$$

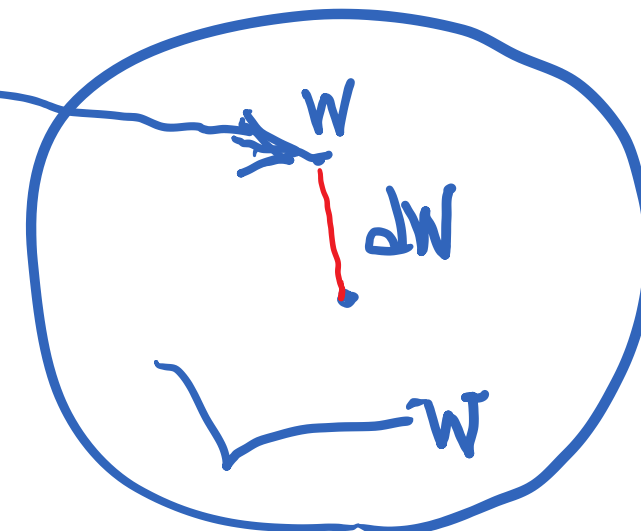
$$p(y, x : W) = c \exp\left\{-\frac{1}{2}(y - \varphi(x; W))^2\right\} q(x)$$

$$\mathbf{F} = E_x[\nabla_W \log p(y, x : W) \nabla_W \log p(y, x : W)]$$

$$\underline{ds^2 = dW^T \mathbf{F} dW}$$

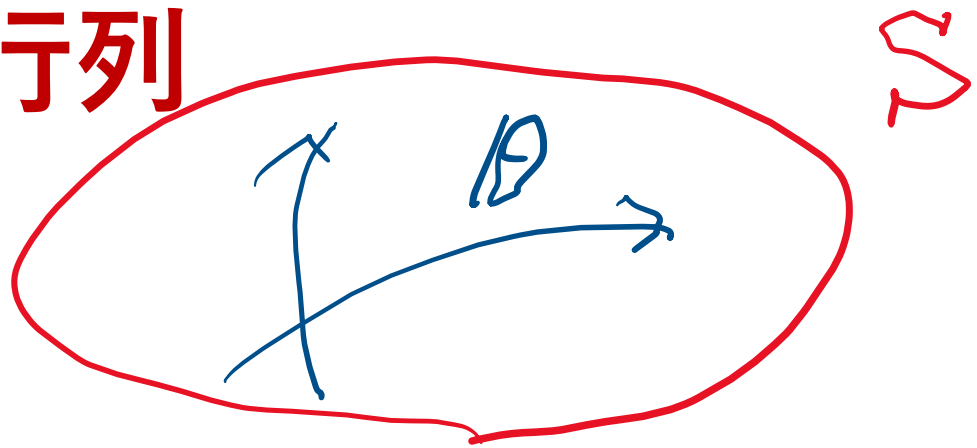
error

Fisher information



Riemannian

確率モデルとFisher情報行列



$$S = \{\theta\} : p(y|x; \theta) = \frac{1}{\sigma} \exp \left\{ -\frac{1}{2} \left(y - f(x, \theta) \right)^2 \right\}$$

$$l = \log p(x, y, \theta) = -\frac{1}{2} \left\{ y - f(x, y, \theta) \right\}^2 - \frac{1}{2} \sigma^2 \quad \text{Cramer-Rao}$$

$$F_{ij}(\theta) = E \left[\frac{\partial}{\partial \theta_i} l \frac{\partial}{\partial \theta_j} l \right] = -E \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} l \right] \quad \checkmark$$

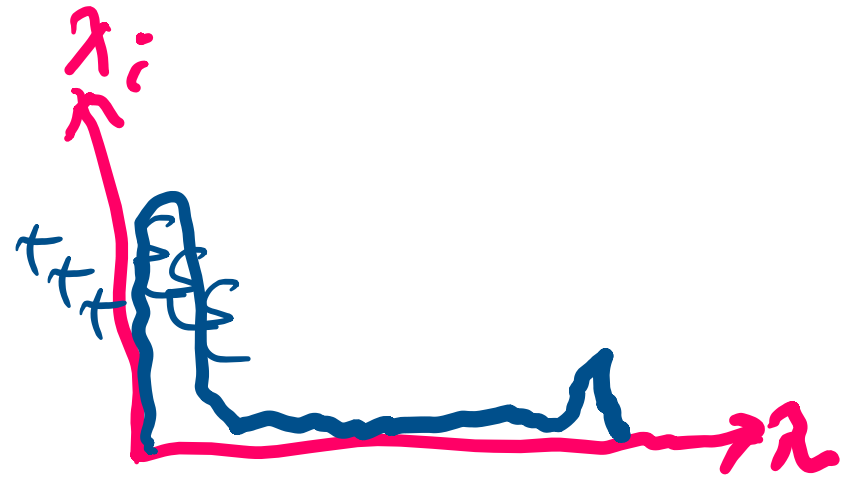
$$|d\theta|^2 = d\theta^T F_{\theta} d\theta \quad \text{リマン計量} \quad (\hat{\theta} - \theta)^2 \leq \frac{1}{n} F^{-1}$$

Fの固有値の分布

$$z = \frac{\partial}{\partial \theta} s(x, y, \theta)$$

$$F = E[z^T z]$$

$$\Delta \theta = -\eta F^{-1} \frac{\partial}{\partial \theta} l$$

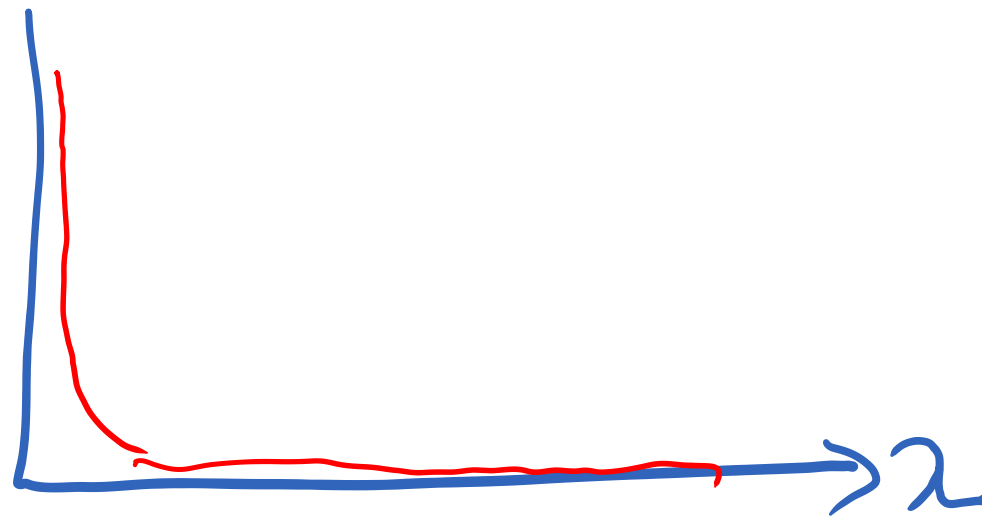


自然勾配降下法 — リーマン空間での再急降下法

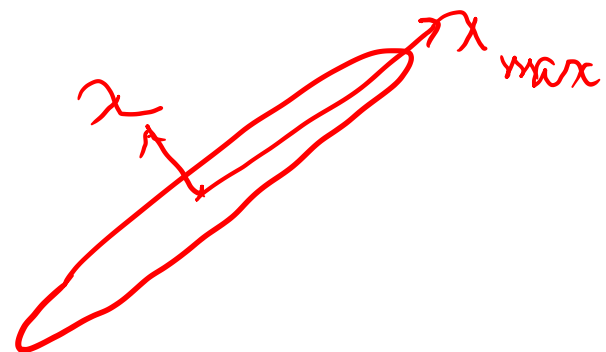
唐木田 theory

eigenvalues of $\tilde{F}$

$$\sum \lambda_i = \frac{1}{n}, \quad \sum \lambda_i^2 = O(1)$$



distorted Riemannian metric



自然勾配法: リーマン空間での最急降下

$$\max \quad dl = l(\boldsymbol{\theta} + d\boldsymbol{\theta}) - l(\boldsymbol{\theta})$$

$$|d\boldsymbol{\theta}|^2 = \varepsilon \quad \text{KL}[p(\mathbf{x}, \boldsymbol{\theta}) : p(\mathbf{x}, \boldsymbol{\theta} + d\boldsymbol{\theta})] = \varepsilon$$

$$\nabla l = \mathbf{F}^{-1}(\boldsymbol{\theta}) \nabla l$$

$$\Delta l \approx -\eta \underbrace{\mathbf{F}^{-1}(\boldsymbol{\theta}) \frac{\partial l}{\partial \boldsymbol{\theta}}}_{\tilde{\nabla} l}$$

$$\tilde{\nabla} l$$

大きくとる!

$$\Delta \boldsymbol{\theta}_t = -\eta_t \tilde{\nabla} l(x_t, y_t; \boldsymbol{\theta}_t)$$

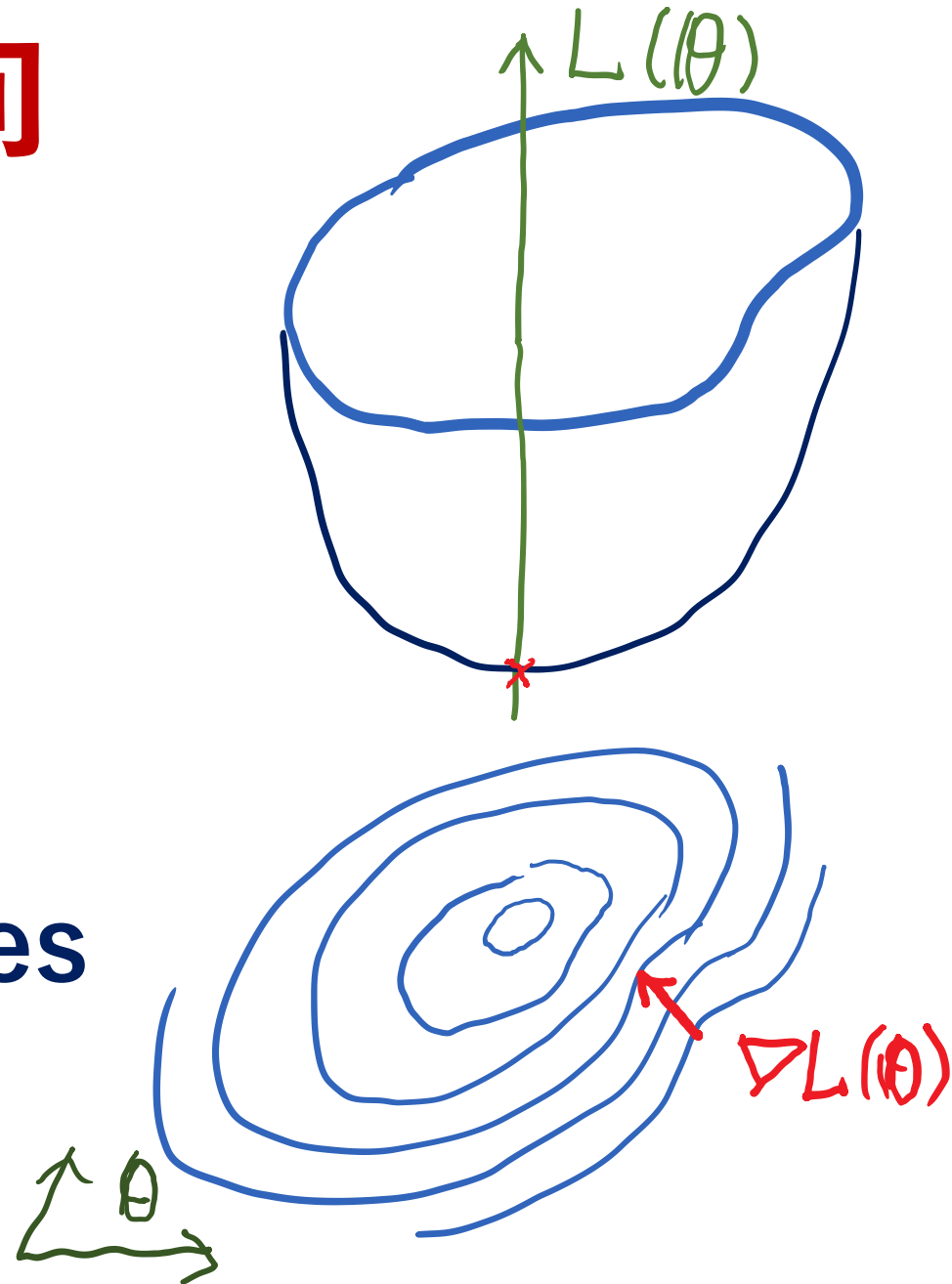
$\mathbf{F}^{-1} \nabla$

$$\Delta \theta_t^i = (1 - \eta \lambda_i) \Delta \theta_{t-1}^i + \varepsilon$$

Lの最急降下方向

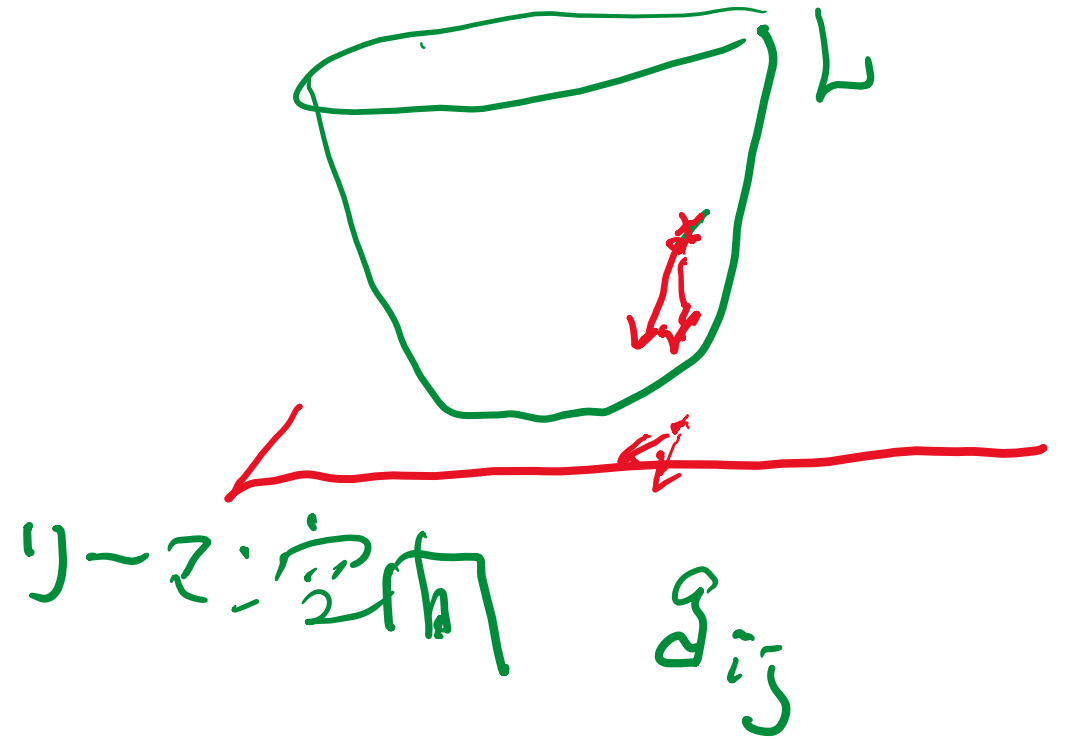
$$\nabla L(\boldsymbol{\theta}) = \left(\frac{\partial L}{\partial \theta_1}, \frac{\partial L}{\partial \theta_2}, \dots, \frac{\partial L}{\partial \theta_n} \right)$$

勾配は最急方向か？
Orthonormal coordinates
in a Euclidean space



Lの最急降下方向

$$\min_a |\nabla \ell a|^2$$
$$a: g_{ij} a^i a^j = \varepsilon^2$$



$$|ds^2| = \sum g_{ij} a^i a^j$$

自然勾配法 Natural gradient Riemannian gradient

$$\min L = \frac{1}{2} \{y - f(x, \theta + \Delta\theta)\}^2 : \quad KL[p(x, \theta) : p(x, \theta + \Delta\theta)] \leq \varepsilon^2$$

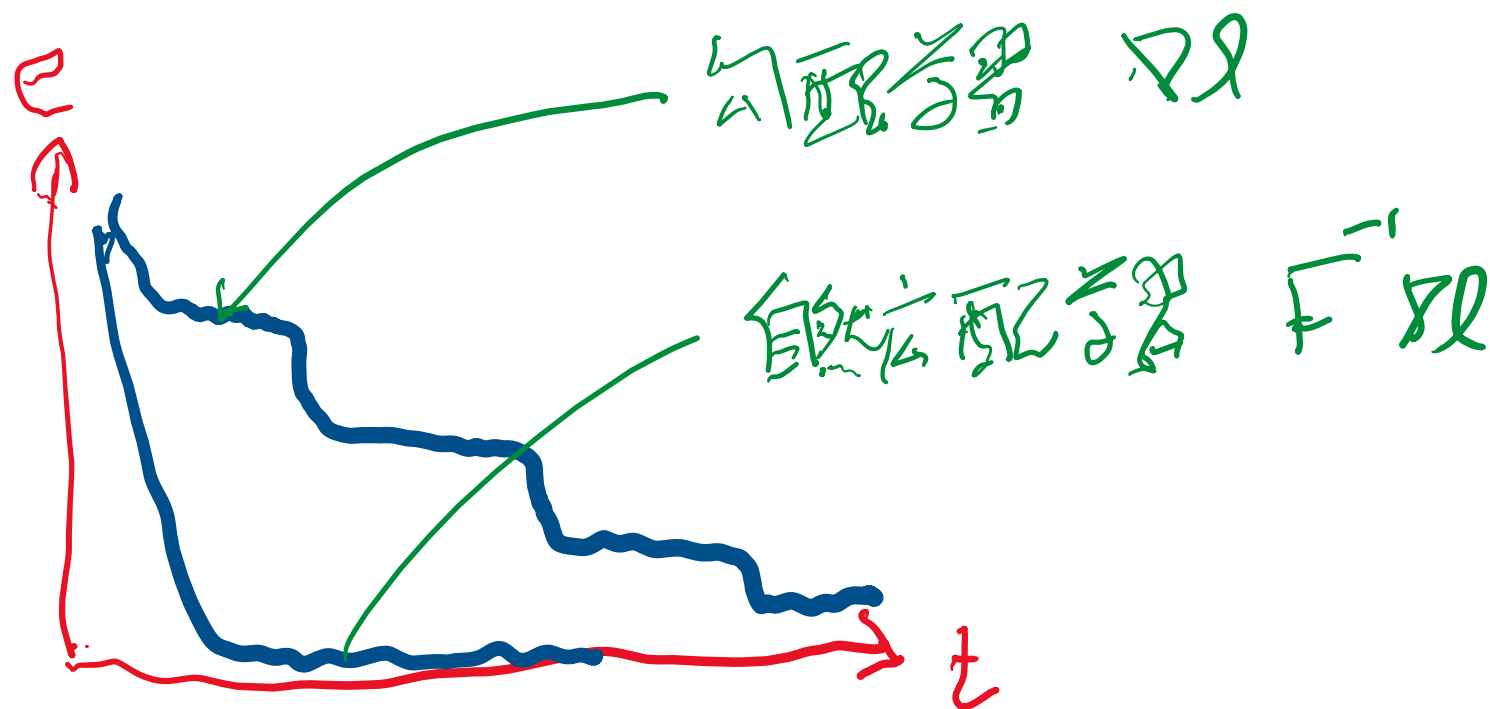
$$\Delta\theta = -\eta \nabla l = -\eta F^{-1}(\theta) \nabla l$$

$$F = E[\nabla l \nabla l] = E[\partial_i \underbrace{\log p(x, \theta)}_f \partial_j \underbrace{\log p(x, \theta)}_f]$$

$$l = \log p(x, y, \theta)$$

Fisher efficient

學習曲線



$$\Delta \theta = -\eta F^{-1} \partial_{\theta} l$$

$$F = \begin{bmatrix} \square & & & & \\ & \square & & & \\ & & \square & & \\ & & & \square & \\ & & & & \square \end{bmatrix} + O\left(\frac{1}{\sqrt{p}}\right)$$

$$G^{-1} = \begin{bmatrix} \square & \\ & \square \end{bmatrix} + O\left(\frac{1}{\sqrt{p}}\right) \quad \times$$

$$F^{-1} F = I + O\left(\frac{1}{\sqrt{p}}\right)$$

Fisher Information

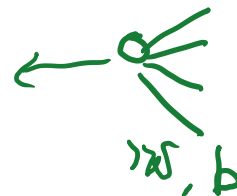
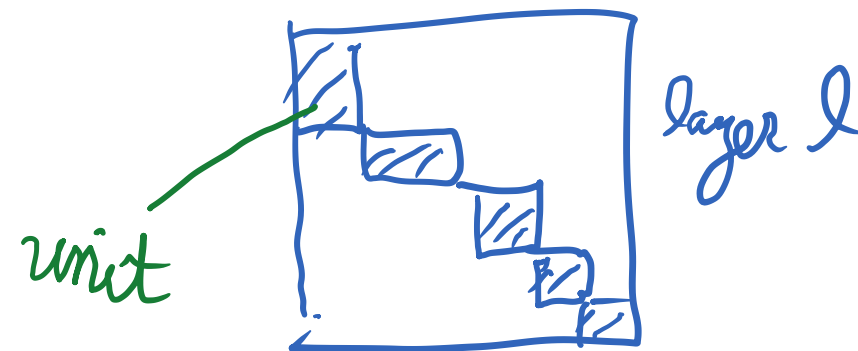
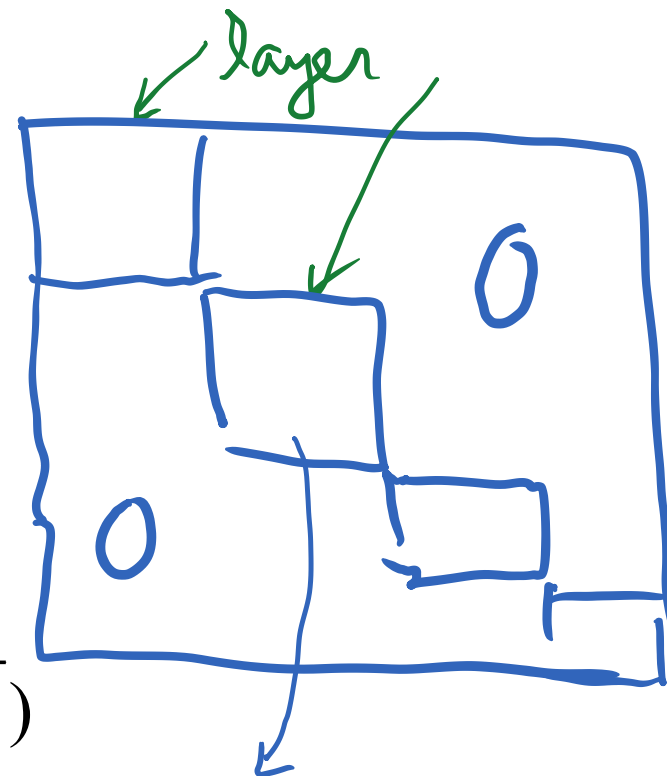
$$\mathcal{G} = E_x \left[\frac{\partial \phi}{\partial W_m} \frac{\partial \phi}{\partial W_l} \right]$$

$$\frac{\partial \phi^l}{\partial W_m} = \phi' W \frac{\partial \phi^{l-1}}{\partial W_m} = B \frac{\partial \phi^{l-1}}{\partial W_m} = BB \dots B \frac{\partial \phi^{m+1}}{\partial W_m}$$

$$\mathcal{G}(W_l, W_m) = \prod \chi_1 E_x \left[\phi' \begin{pmatrix} l \\ \mathbf{w}_i \end{pmatrix}^2 \begin{matrix} l-1 & l-1 \\ \mathbf{x} & \mathbf{x} \end{matrix} \right] + O_p(1/\sqrt{n})$$

$$\mathcal{G}(W_l, W_m) = 0 \sim O_p(1/\sqrt{n}), \quad l \neq m$$

$$\mathcal{G} \begin{pmatrix} l & l \\ \mathbf{w}_i & \mathbf{w}_j \end{pmatrix} = 0 \sim O_p(1/\sqrt{n}), \quad i \neq j$$



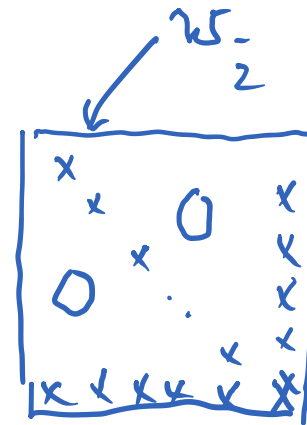
$$F^{-1}$$

Yann Ollivier

Block-wise diagonal
Unitwise diagonal
Bias+diagonal

KFAC
Arata

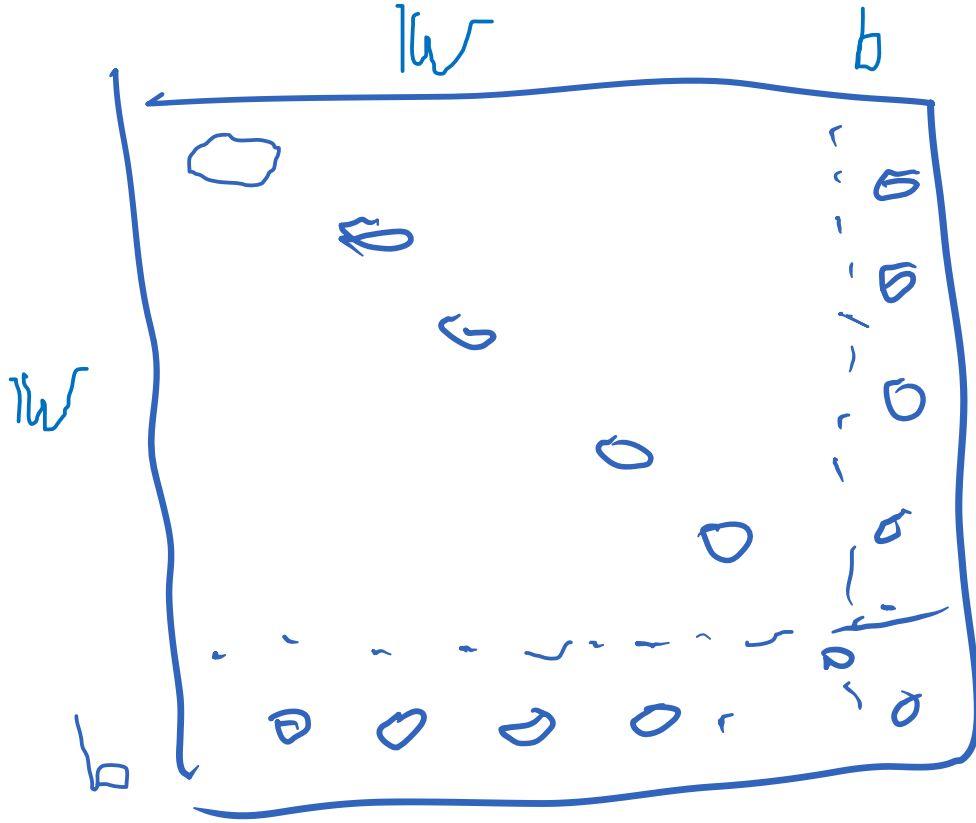
$$F^{-1} = \begin{array}{|c|} \hline \text{[diagonal blocks]} \\ \hline \end{array} \quad O(1/\sqrt{n})$$



$$\text{bias} + C W^T W$$

(Gaussian input)

G=



+ WW

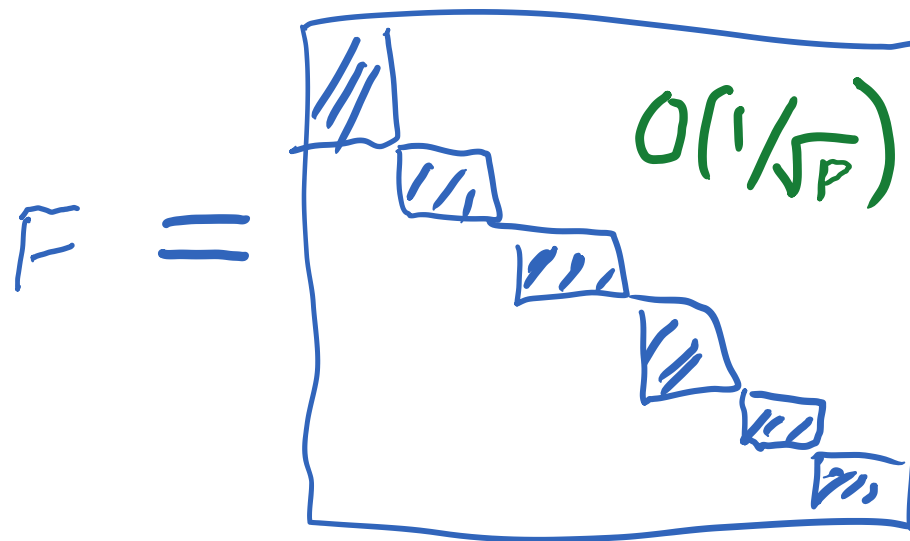
Ollivier, Marceau-Caron : quasi-diagonal natural gradient

Fisher 情報行列 — 大規模の場合

Amari et al., ISTATS, 2019

Unitwise diagonal

F-inv: no problem!!



Goodnews and bad news

G^* : unitwise-diagonal matrix

$$G \rightarrow G^* : n \rightarrow \infty$$

$$G^{-1} \not\rightarrow G^{*-1} : n \rightarrow \infty$$

シミュレーション:

Natural gradient is no more dream!

TANGO: Ollivier
recursive formula for

$$\mathbf{F}^{-1} \nabla l$$

Neural tangent kernel

經驗自然勾配法

$$\Delta\theta = -\eta \hat{F}^\dagger \nabla l = -\eta X^T K^{-1} e$$

G. Zhang, J. Martens and R. Grosse, 2019
T. Cai et al. 2019

Empirical Fisher information

Empirical Fisher information

$$\nabla_{\theta} f(x_1, \theta), \nabla_{\theta} f(x_2, \theta), \dots, \nabla_{\theta} f(x_n, \theta)]$$

$$\frac{1}{n} \sum \nabla_{\theta} f(x_s, \theta) \nabla_{\theta} f(x_s, \theta)^T : \text{Fisher}$$

$$\sum_{\theta} \nabla_{\theta} f(x_s, \theta) \nabla_{\theta} f(x_t, \theta)^T : \text{NT kernel}$$

$$-\eta X^T K^{-1} e$$

$$X = \begin{bmatrix} x_1 & \dots & x_n \\ \theta_1 & \dots & \theta_p \end{bmatrix}$$

$$X^T = \begin{bmatrix} x_1 & \dots & x_n \\ \theta_1 & \dots & \theta_p \end{bmatrix}$$

$$F(\theta) = \frac{1}{n} \sum \nabla_{\theta} f(x_s, \theta) \nabla_{\theta} f(x_s, \theta)^T : \text{Fisher}$$

$$K(x, x') = XX^T = \sum_{\theta} \nabla_{\theta} f(x_s, \theta) \nabla_{\theta} f(x_t, \theta)^T : \text{NT kernel}$$

$$\hat{F}^{\dagger} = nX^T (XX^T)^{-1}$$

$$\Delta \theta = -\eta \hat{F}^{\dagger} \nabla l = -\eta X^T K^{-1} e$$

$$X = \begin{bmatrix} x_1 & \dots & x_n \\ \theta_1 & \dots & \theta_p \end{bmatrix}$$

$$X^T = \begin{bmatrix} x_1 & \dots & x_n \\ \theta_1 & \dots & \theta_p \end{bmatrix}$$

G. Zhang, J. Martens and R. Grosse, 2019

T. Cai et al. 2019

Karakida & Osawa, 2020

神經接核理論

(Neural tangent kernel)

Jacot et al; Neural tangent kernel (2018)

学習の力学

Parameter space M

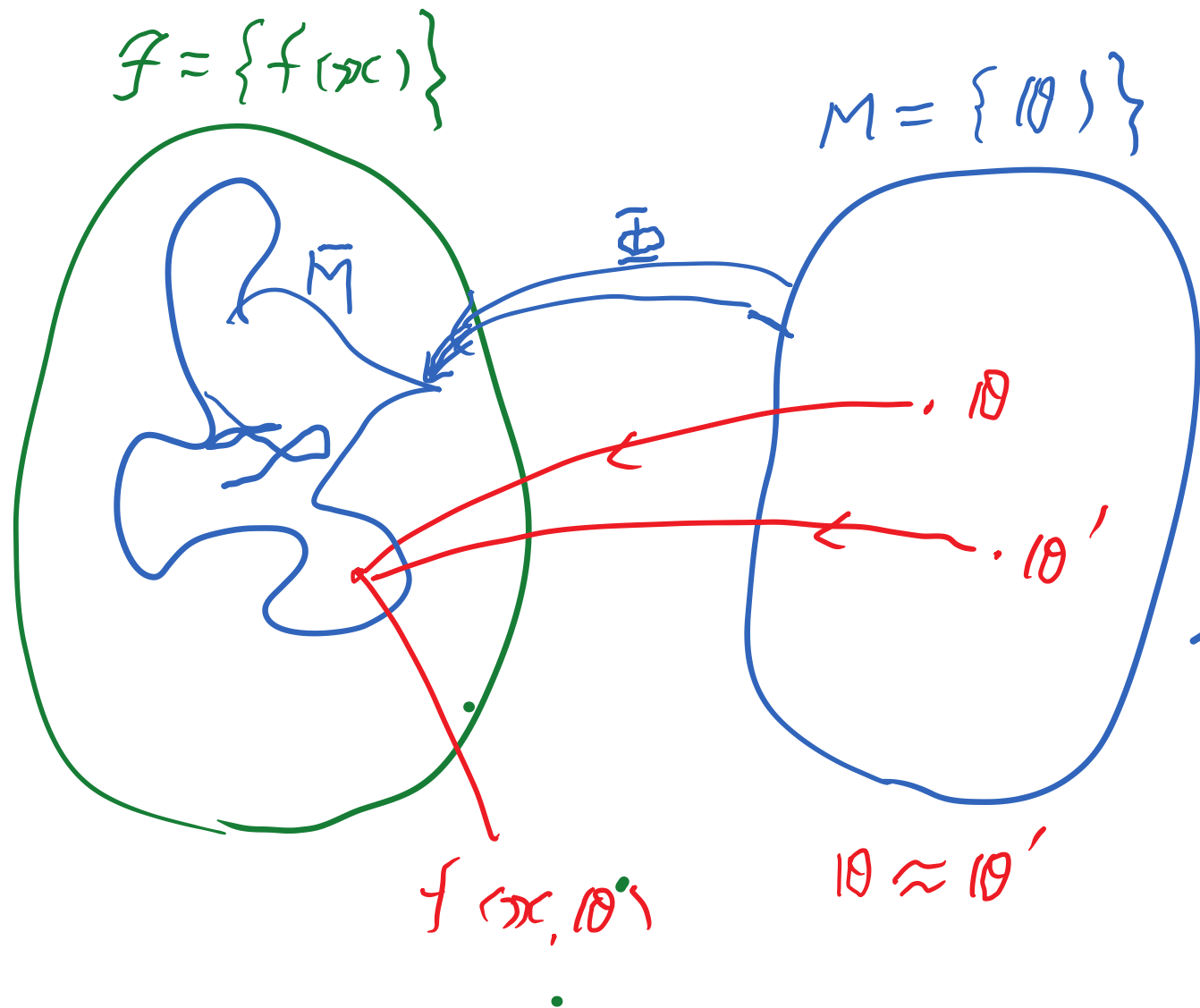
Function space $\mathcal{F}$

$$\theta = \theta' \Leftrightarrow f(x, \theta) = f(x, \theta')$$

$$M \Rightarrow \bar{M} = \Phi(M) \subset \mathcal{F}$$

$$\bar{M} = \Phi(M) / \approx$$

Equivalence relation
Singular structure



学習：関数空間とパラメータ空間

$$y = f(x, \theta)$$

NTK : neural tangent kernel
神経接核理論

$$\boxed{f} \Leftrightarrow \boxed{\theta}$$

$$\mathcal{F} = \{f(x)\}$$

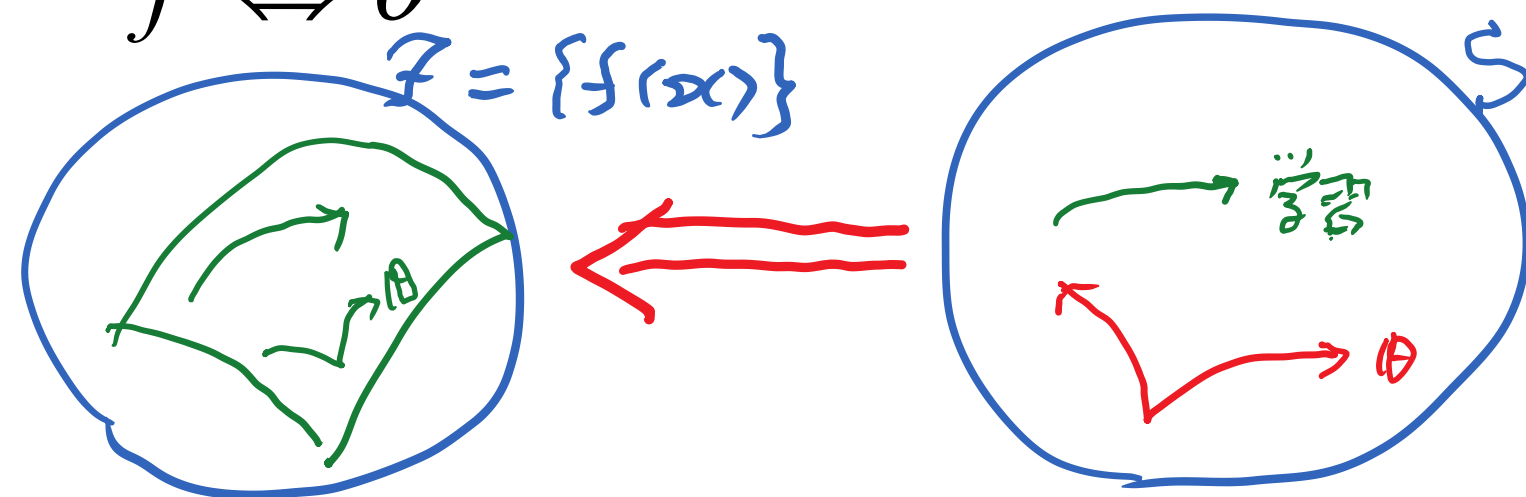
$$\mathcal{S} = \{\theta\}$$

同値類

構造的

$$f(x, \theta) = f(x, \theta')$$

$$\theta \neq \theta'$$



Jacot et al; Neural tangent kernel (2018)

$$y = f(x, \theta); \quad l(x, \theta) = \frac{1}{2} (y - f(x, \theta))^2$$

$$l(x, \theta) = \frac{1}{2} e^2; \quad e = f(x, \theta) - f(x, \theta^*)$$

$$\partial_t \theta = -\eta \partial_\theta f(x') (f(x', \theta) - f(x', \theta^*))$$

$$\partial_t f(x, \theta) = \partial_\theta f(x) \partial_t \theta = -\eta \partial_\theta f(x) \cdot \partial_\theta f(x') e(x', \theta)$$

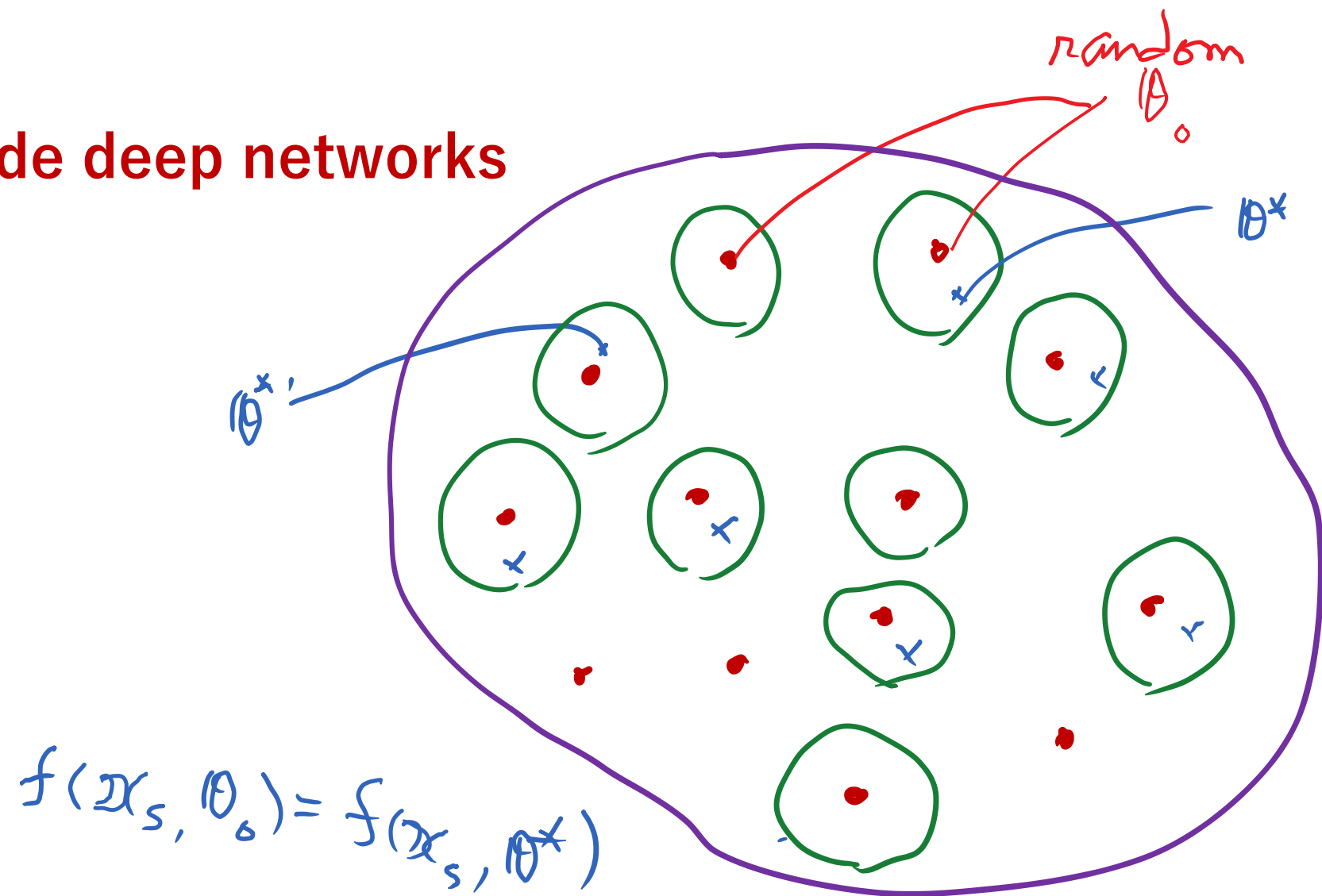
$$K(x, x'; \theta) = \partial_\theta f(x) \cdot \partial_\theta f(x')$$

$$\partial_t f(x, \theta) = -\eta \langle K(x, x'; \theta) e(x', \theta) \rangle$$

K; Gaussian kernel

どんな関数もランダム回路の近傍で見 つかる

- Sufficiently wide deep networks



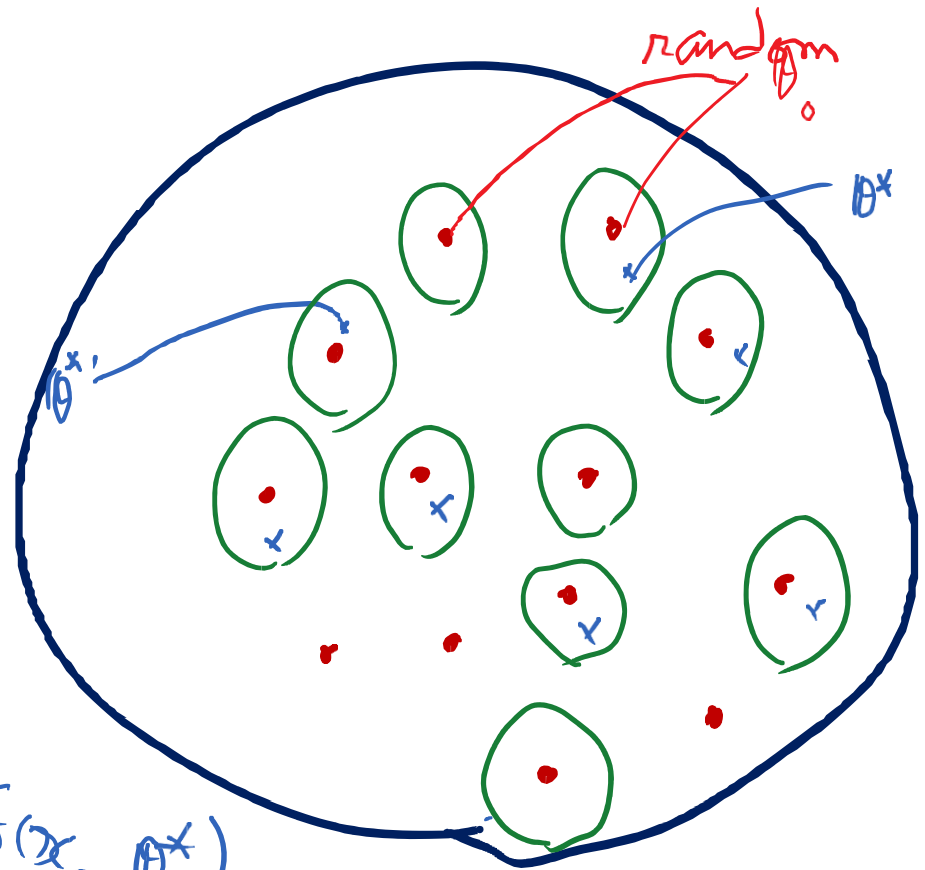
Training data $n \ll p$ $D = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$

$$\theta^* = \arg \min \frac{1}{2} \sum_{s=1}^n \{y_s - f(\mathbf{x}_s, \theta)\}^2$$

定理 どんなランダム回路
でもその微小近傍正解があ
る。

$$|\theta_0 - \theta^*|^2 \approx O\left(\frac{1}{p}\right)$$

$$f(\mathbf{x}_s, \theta_0) = f(\mathbf{x}_s, \theta^*)$$

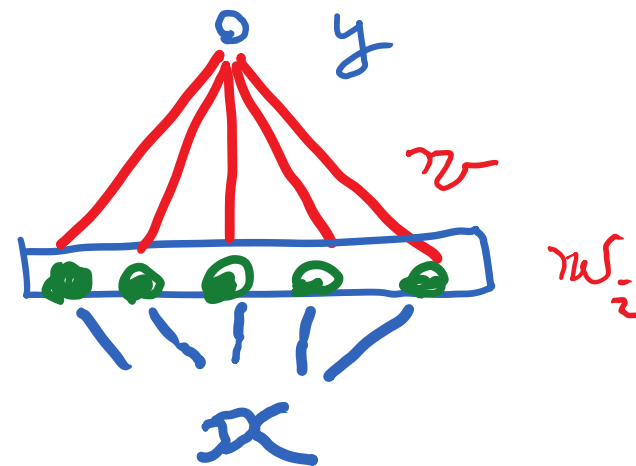


単純パーセプトロンの場合

$$y = \sum v_i \phi(\mathbf{w}_i \cdot \mathbf{x})$$

$\mathbf{w}_i \sim \text{random}; \text{fixed}$

$$v_i \sim N(0, \frac{\sigma^2}{p}), \quad \sigma^2 = 1$$



$$y_s^* = \sum (v_i + \Delta v_i) \phi(\mathbf{w}_i \cdot \mathbf{x}_s), \quad s = 1, \dots, n$$

correction

$$e_s = \sum v_i \phi(\mathbf{w}_i \cdot \mathbf{x}) - y_s^*$$

$$X_{si} = \phi(\mathbf{w}_i \cdot \mathbf{x}_s) = \frac{\partial f(\mathbf{x}_s, \theta)}{\partial \theta}$$

$$\mathbf{e} = X \Delta \mathbf{v}$$

$$\Delta \mathbf{v} = X^\dagger \mathbf{e}$$

$$\begin{bmatrix} \Delta v \end{bmatrix} = \begin{matrix} n \\ p \end{matrix} \begin{bmatrix} X^\dagger \end{bmatrix} \begin{bmatrix} e \end{bmatrix}$$

$$\begin{bmatrix} e \end{bmatrix} = n \begin{matrix} p \\ X_{si} \end{matrix} \begin{bmatrix} v \end{bmatrix}$$

$$X^\dagger = X^T (XX^T)^{-1}$$

$$K = XX^T: \text{ tangent kernel } (K_{st} = K(\mathbf{x}_s, \mathbf{x}_t))$$

$$\square \\ f = -\eta Ke = -\eta \partial_\theta f(x) \cdot \partial_\theta f(x') e$$

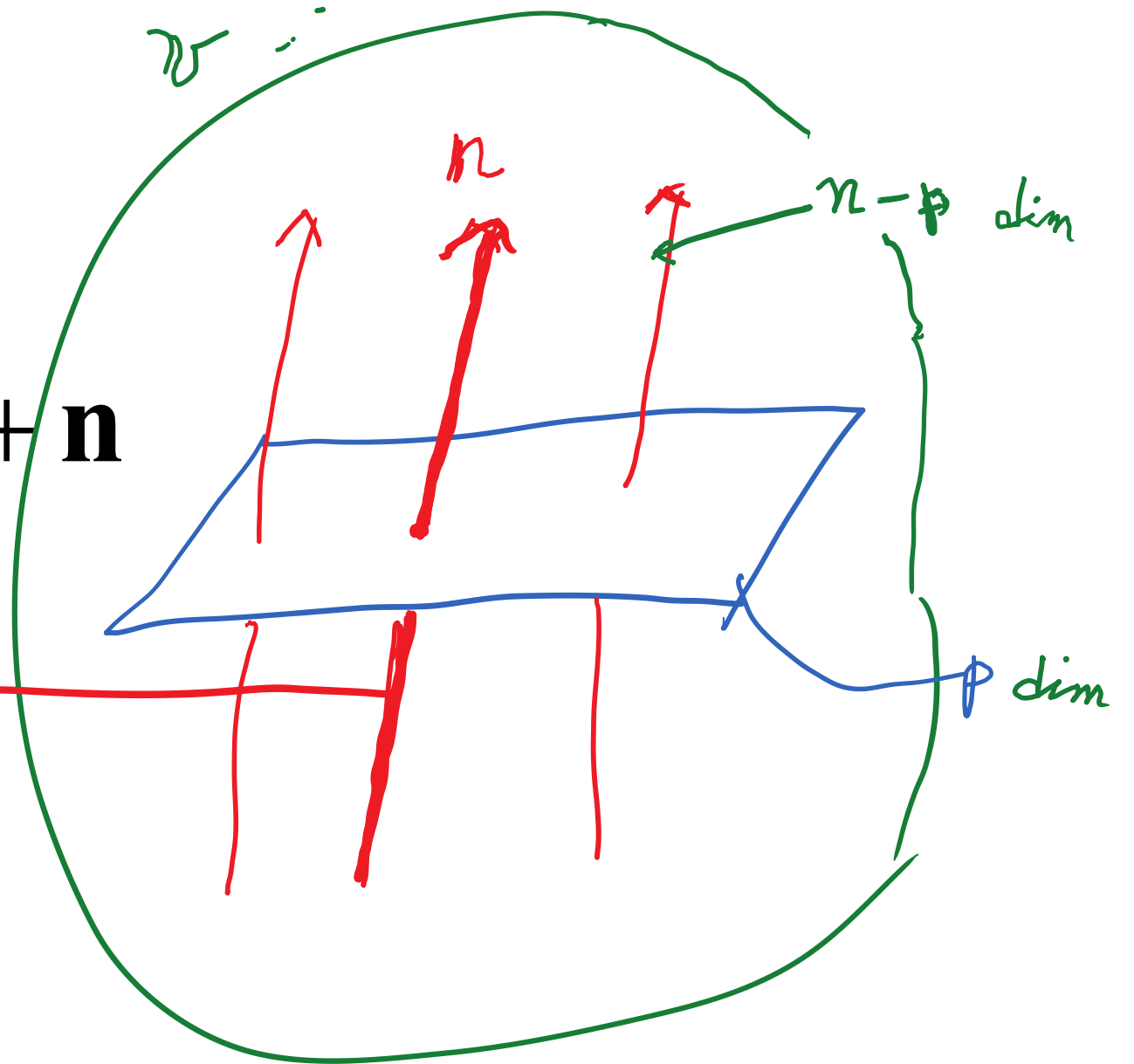
$$\boxed{K^{-1} \mathbf{e} \square O\left(\frac{1}{p}\right); \quad \Delta \mathbf{v} \square O\left(\frac{1}{p}\right)} \quad K_{st} = \sum \varphi(\mathbf{w}_i \cdot \mathbf{x}_s) \varphi(\mathbf{w}_i \cdot \mathbf{x}_t) \\ = p \mathbb{E}[\varphi(\mathbf{w}_i \cdot \mathbf{x}_s) \varphi(\mathbf{w}_i \cdot \mathbf{x}_t)]$$

$$\mathbf{e} = X \Delta \mathbf{v}$$

$$N = \{\mathbf{n} \mid X\mathbf{n} = \mathbf{0}\}$$

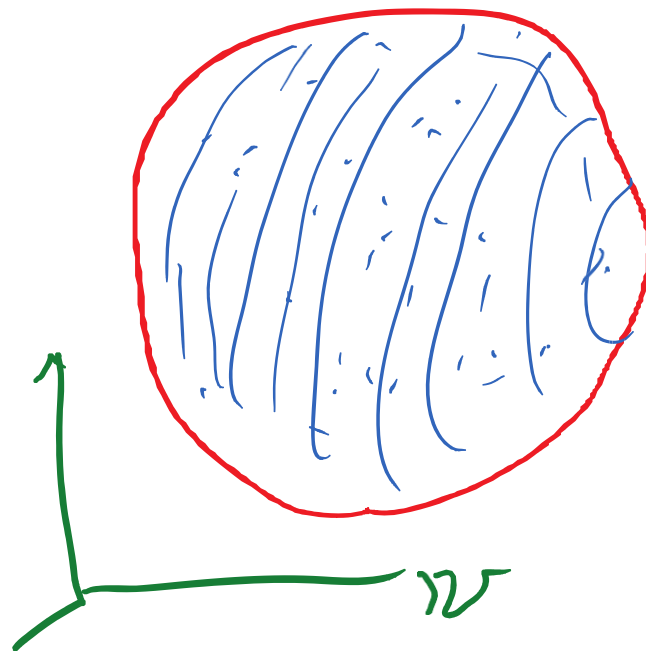
$$\Delta \mathbf{v} = X^\dagger \mathbf{e} + \mathbf{n} = \Delta \mathbf{v}^* + \mathbf{n}$$

$$\Delta \mathcal{V} \approx \Delta \mathcal{V}^* + \mathcal{N}$$



Random initial weights v

$$v_i \sim N(0, \frac{1}{p})$$
$$\sum v_{i_2} = 1$$



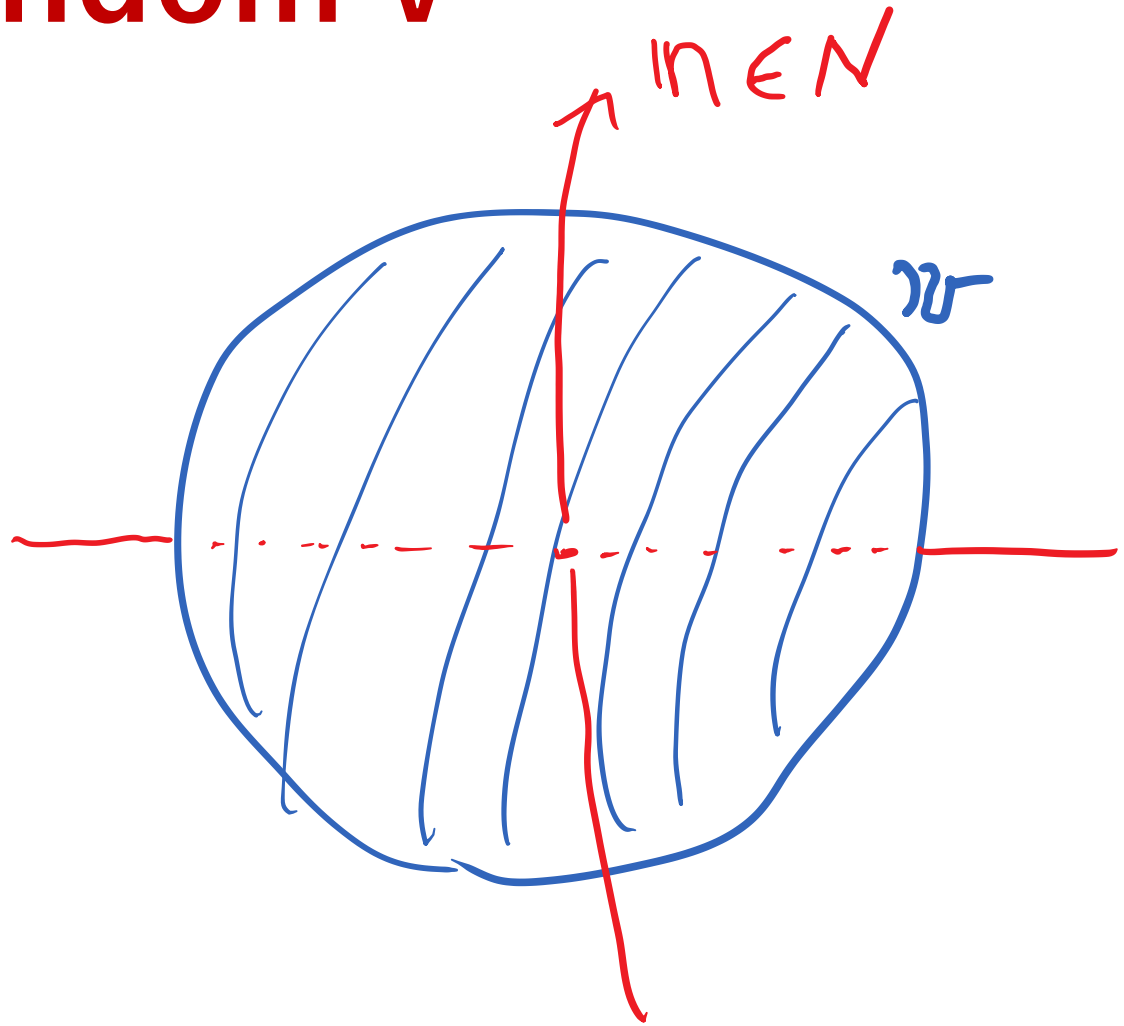
n
↑
equivalent

Distribution of random $\mathbf{v}$

$$v_i \sim M(0, \frac{1}{p})$$

$$|\mathbf{v}|^2 = \sum v_i^2 = 1$$

$$\mathbf{v} \approx \mathbf{v} + \mathbf{n}$$



f^* : target function $f(\mathbf{x}_s)$

$\mathbf{v}^*$: minimal realization

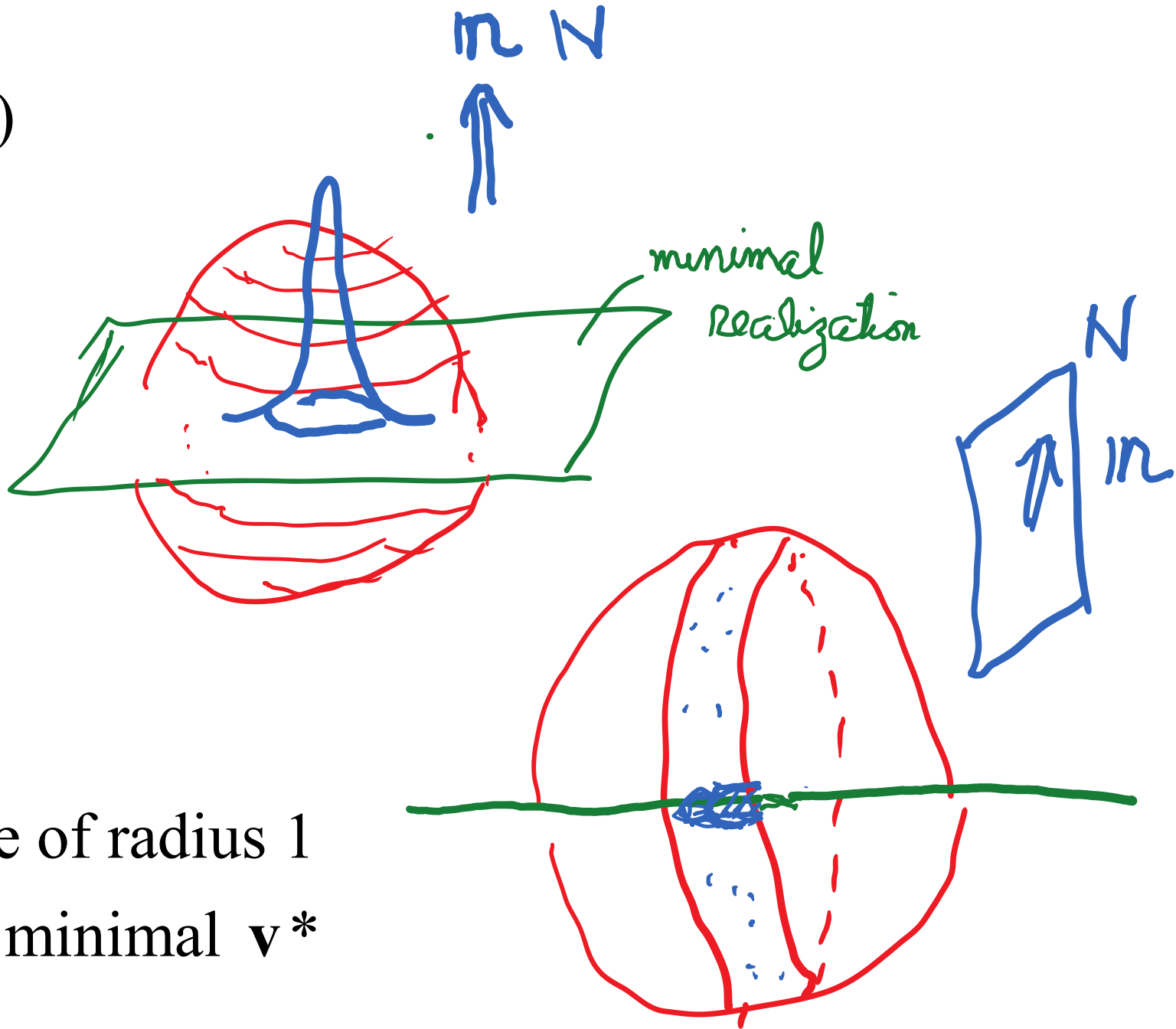
$$f^* = X\mathbf{v}^* = \sum v_i \phi(\mathbf{w}_i \cdot \mathbf{x})$$

$$\mathbf{v}^* = X^\dagger f^*$$

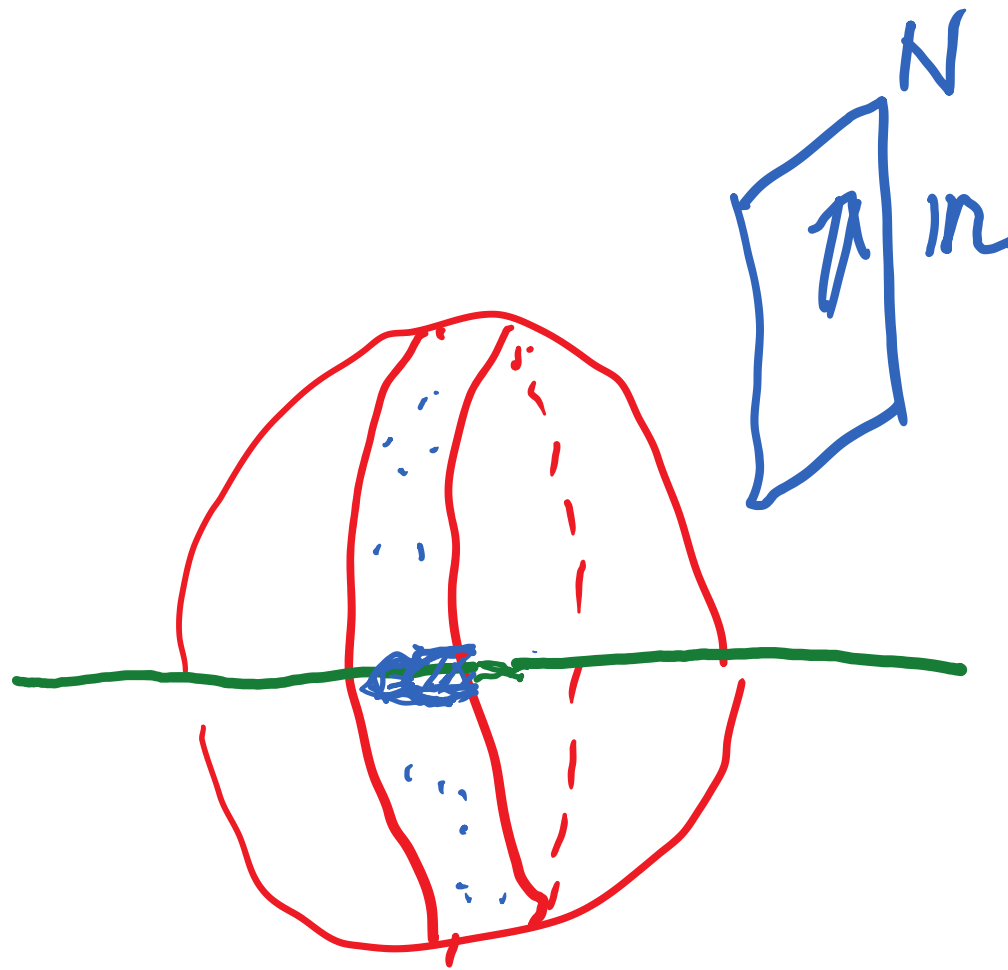
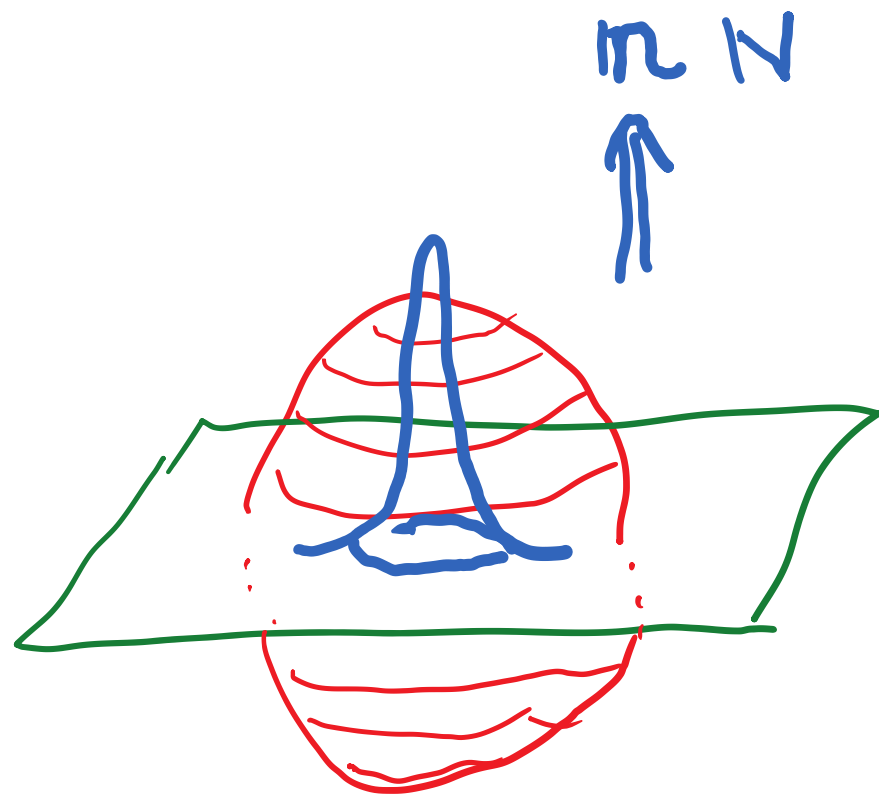
$$\|\mathbf{v}^*\| = O\left(\frac{1}{\sqrt{(p-n)}}\right)$$

$p(\mathbf{v}) \sim$ uniform on sphere of radius 1

$q(\mathbf{v}^*) \sim$ distribution of minimal $\mathbf{v}^*$

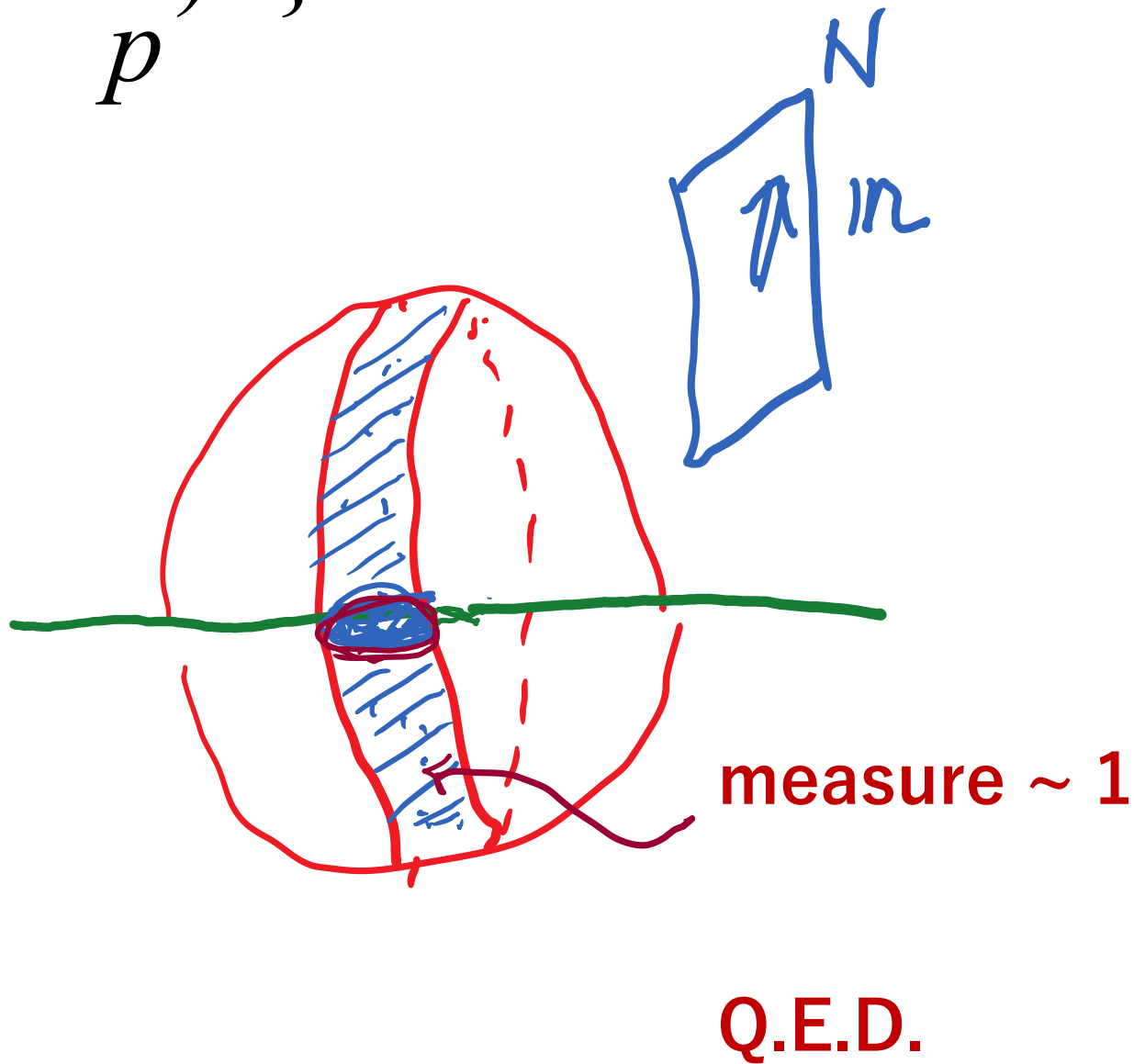
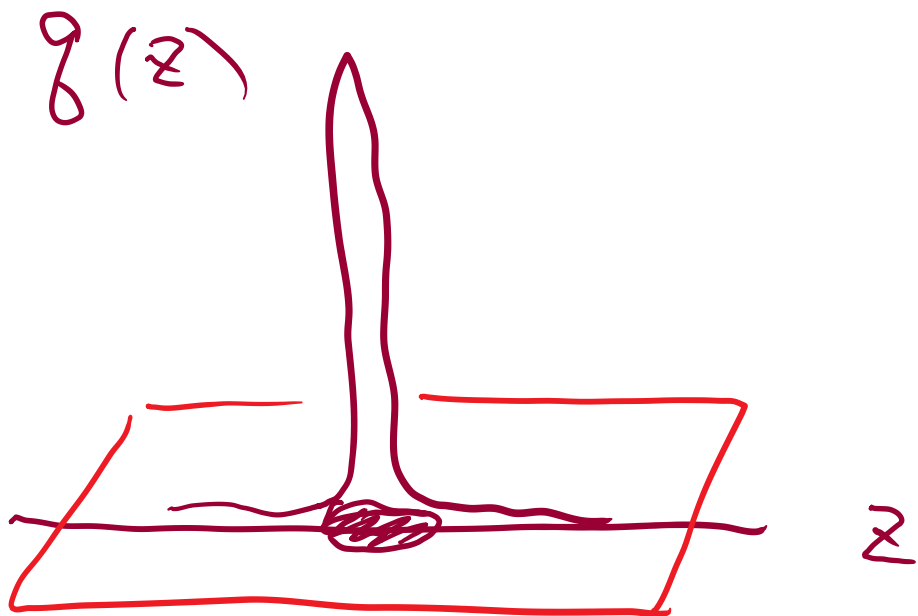


$$q(z) = c \exp \left\{ -\frac{z^2}{2p} \right\}$$



$$q(z) \approx \{\sqrt{(1-z^2)}\}^{p-1} \approx \left\{\left(1-\frac{\varepsilon}{p}\right)^{\frac{p}{\varepsilon}}\right\}^{\frac{\varepsilon}{2}} \approx e^{-\frac{pz}{2}}$$

$$\frac{\varepsilon}{p} = z^2$$



$$\mathbf{e} = X\Delta\mathbf{v}; \quad \Delta\mathbf{v}^* = X^\dagger \mathbf{e} \quad \text{Generalized inverse}$$

$$X^\dagger = X^T (XX^T)^{-1} \quad O\left(\frac{1}{p}\right)$$

$$K = XX^T: \text{ tangent kernel } (K_{st} = K(\mathbf{x}_s, \mathbf{x}_t))$$

$$f = Ke = \partial_\theta f(x) \cdot \partial_\theta f(x') e(x', y')$$

$$K^{-1} \mathbf{e} \begin{matrix} \text{?} \\ \mathcal{O} \end{matrix} \quad O\left(\frac{1}{p}\right); \quad \Delta\mathbf{v}^* \begin{matrix} \text{?} \\ \mathcal{O} \end{matrix} \quad \frac{1}{p}$$

$$\mathbf{v} \sim O\left(\frac{1}{\sqrt{p}}\right)$$

$$XX^T = \sum_{i=1}^p \underbrace{\partial f(x_s)}_{\varphi(w_i \cdot x_s)} \underbrace{\partial f(x_t)}_{O(p)}$$

One layer: both v and w are trainable

$$y = f(\mathbf{x}, \theta) = \sum v_i \phi(\mathbf{w}_i \cdot \mathbf{x})$$

$$\theta = (v_i, \mathbf{w}_i) \quad \Delta f = X \Delta \theta;$$

$$X = \partial_{\theta} f = [\phi(\mathbf{w}_i \cdot \mathbf{x}) \quad v_i \phi'(\mathbf{w}_i \cdot \mathbf{x}) \mathbf{x}] = [X_{is}]$$

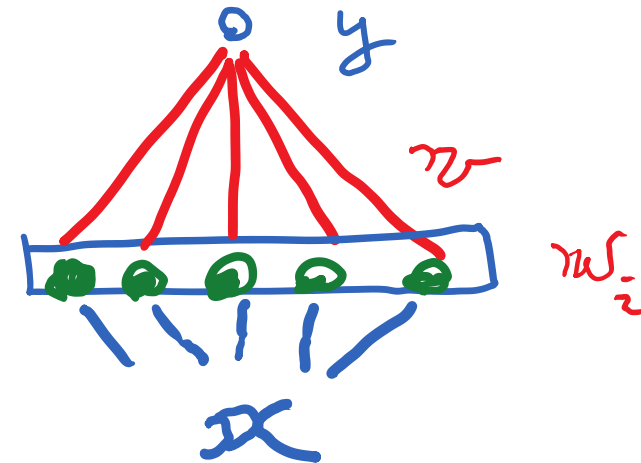
$$\mathbf{f}^* = f(\mathbf{x}_s, \theta) + X \Delta \theta$$

$$\Delta \theta = X^{\dagger} \mathbf{e} \quad X^{\dagger} = X^T (X X^T)^{-1} = X^T K^{-1}$$

$$K = X X^T = \overset{1}{K} + \overset{2}{K} \sim O(p) + O(p);$$

$$\overset{2}{K} = p \mathbb{E}[v_i \phi'(\mathbf{w}_i \cdot \mathbf{x}_s) v_i \phi'(\mathbf{w}_i \cdot \mathbf{x}_t)]$$

$$\Delta \theta = O\left(\frac{1}{p}\right)$$



一般の深層回路の場合

$$y = f(\mathbf{x}, \theta) = \sum_{i=1}^L v_i \phi(\mathbf{w}_i \cdot \mathbf{x}) \quad \theta = \{(v_i, \mathbf{w}_i)\}$$

$$\Delta f = X \Delta \theta; \quad X = \begin{bmatrix} \phi(\mathbf{w}_1 \cdot \mathbf{x}) & v_1 \phi'(\mathbf{w}_1 \cdot \mathbf{x}) & \mathbf{x} & \boxed{} & \boxed{} & \boxed{} & \boxed{} \end{bmatrix}$$

$$\Delta \theta = X^\dagger \mathbf{e} \quad X^\dagger = X^T (X X^T)^{-1}$$

$$K = X X^T = \sum_{l=1}^{L+1} K^l$$

Tangent kernel; Jacot, Lee
Recursively calculated
(mean field approx.)

$$\Delta \theta = O\left(\frac{1}{p}\right) = O\left(\frac{n}{Lp}\right)$$

Jacot et al; **Neural tangent kernel**

$$y = f(x, \theta); \quad l(x, \theta) = \frac{1}{2} (y - f(x, \theta))^2$$

$$l(x, \theta) = \frac{1}{2} \{y - f(x, \theta)\}^2; \quad e = f(x, \theta) - f(x, \theta^*)$$

$$\partial_t \theta = -\eta \partial_\theta f(x') (f(x', \theta) - f(x', \theta^*))$$

$$\partial_t f(x, \theta) = \partial_\theta f(x, \theta) \partial_t \theta = -\eta \partial_\theta f(x) \cdot \partial_\theta f(x') e(x', \theta)$$

true parameter

training examples

K; Gaussian kernel

$$K(x, x'; \theta) = \partial_\theta f(x, \theta) \cdot \partial_\theta f(x', \theta)$$

$$\partial_t f(x, \theta) = -\eta \langle K(x, x'; \theta) e(x', \theta) \rangle$$

f - y

General Deep Network

$$X = [X^{L+1} X^L \cdots X^l \cdots X^1]$$

$$\Delta\theta = X^\dagger \mathbf{e} \quad X^\dagger = X^T (XX^T)^{-1}$$

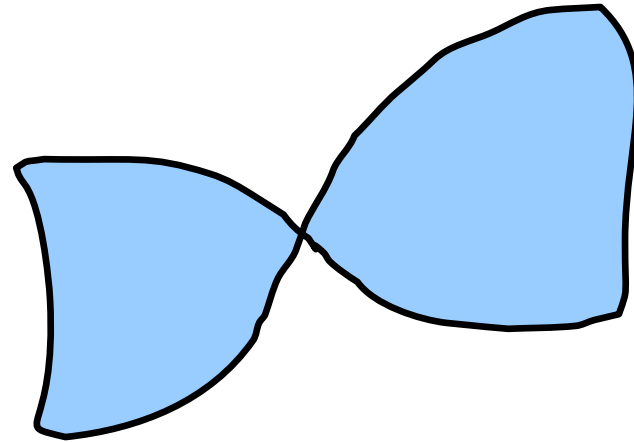
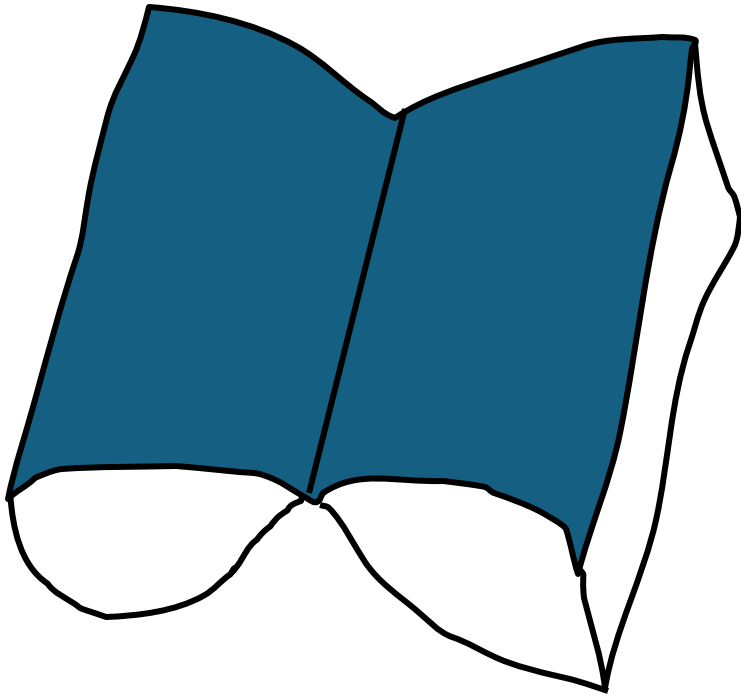
$$K = XX^T = [K^{L+1} K^L K^{L-1} \cdots K^l \cdots K^1]$$

$$\Delta\theta = O\left(\frac{n}{pL}\right)$$

Tangent kernel
Jacot, Lee
Recursively calculated
(mean field approx.)

パラメータ空間の特異構造

特異点 singularities



同値類と商空間 quotient space singularity

$$\theta \approx \theta' \Leftrightarrow f(x, \theta) = f(x, \theta')$$

$$F = M / \approx$$

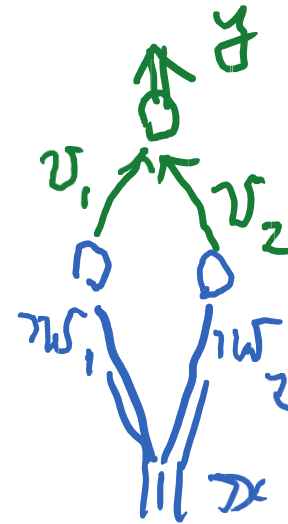
基本特異構造

$$y = v_1 \varphi(w_1 \cdot x) + v_2 \varphi(w_2 \cdot x)$$

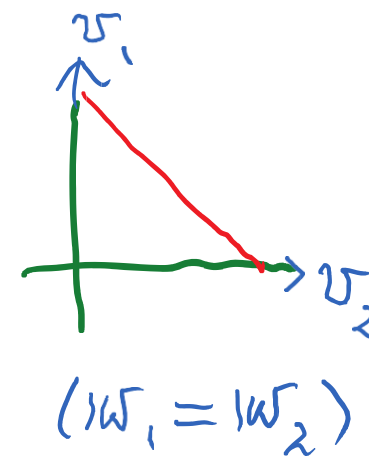
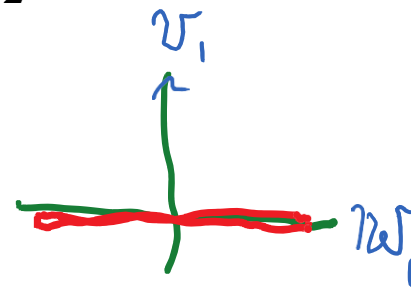
$$v_1 = 0 \Rightarrow w_1 \approx w'_1$$

$$w_1 = w_2 \Rightarrow (v_1, v_2) \approx (v'_1, v'_2),$$

$$v_1 + v_2 = v'_1 + v'_2$$



線形部分空間の和集合



Fisher 情報行列の縮退

$$|F| \approx 0$$

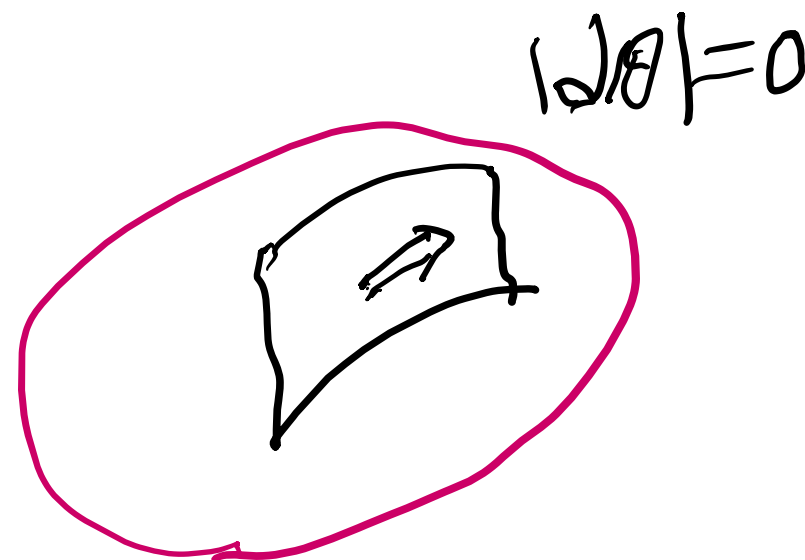
$$d\theta^T H d\theta = |d\theta|^2 = 0$$

Cramer-Rao定理

同定不能

同値

$$E[ee^T] \approx \frac{1}{n} F^{-1} \quad \checkmark$$



学習の遅滞

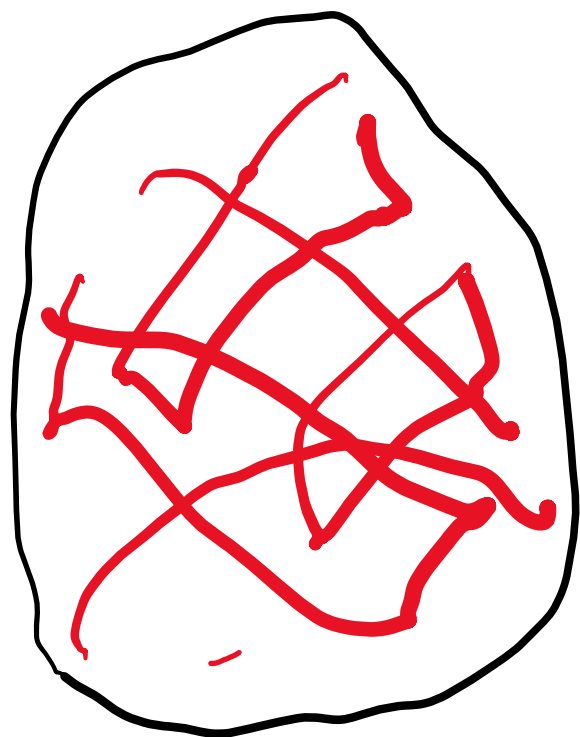


自然勾配学習

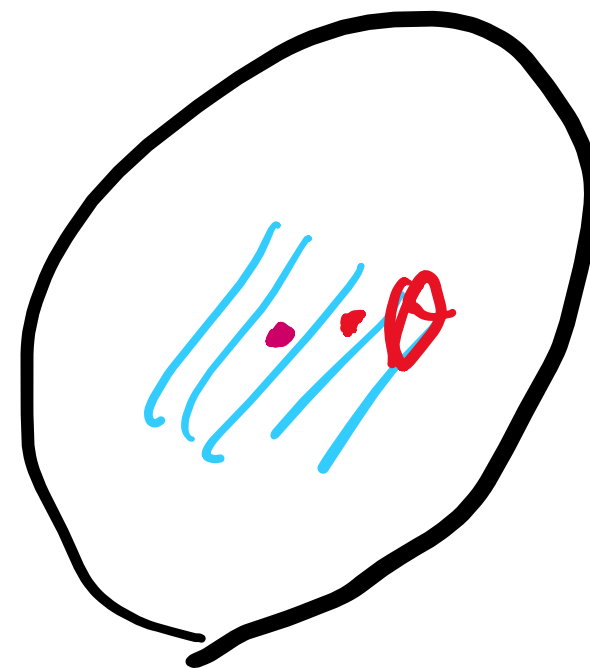


代数幾何——特異点のある統計モデルにおける推論

渡辺澄夫



$$|e e^T| \approx \frac{1}{2(n)} F^{-1}$$



特異部分多様体の網の目

•

理論の貧困

多きいことは無限にいいことか？

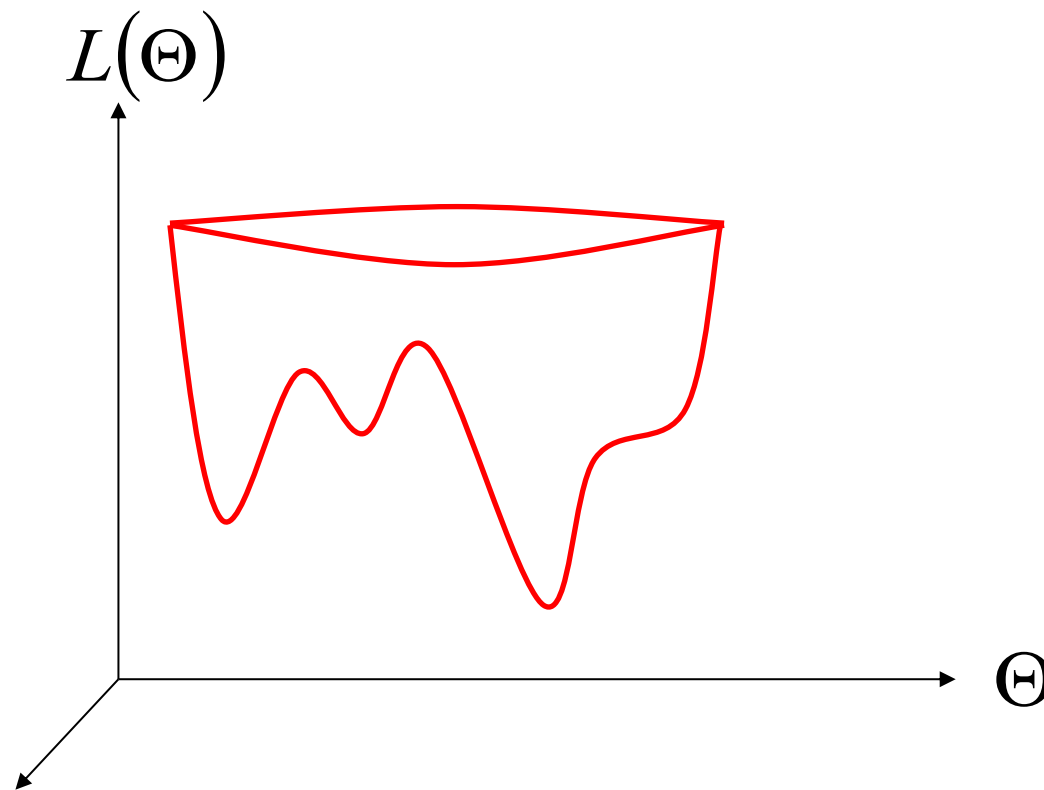
スケール則



創発

大域解と局所解

Simulated annealing
Quantum annealing



汎化誤差と訓練誤差

N: number of examples

P : number of parameters

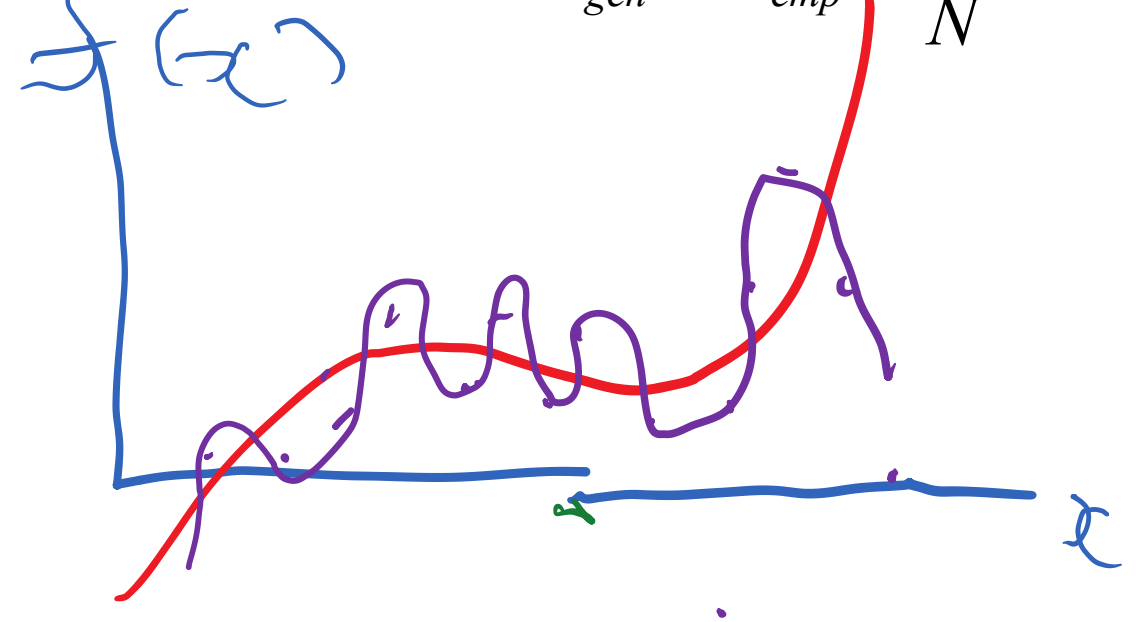
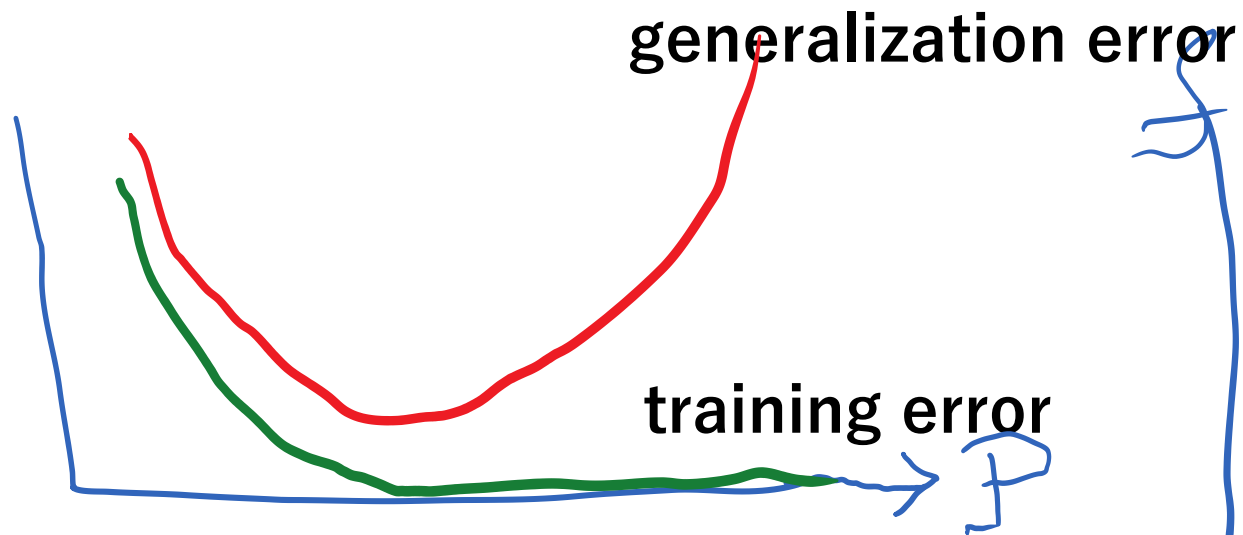
$$y = f(x, \theta) + \varepsilon$$

過学習Overfitting

$$L_{emp} = \frac{1}{N} \sum |y_i - f(x_i, \theta)|^2$$

$$L_{gen} = E[|y - f(x, \theta)|^2]$$

$$L_{gen} \approx L_{emp} + \frac{P}{N}$$



汎化誤差の二重降下

Belkin, Hsu, Xu; 2018

— error curve: **double descent**

